Measured Build Summary

RTL inference samples: 10000 accuracy: 0.9551 exact score match: True

On-chip learning samples: 10000 updates: 25 score mismatches: 0

Activity gate: abs(feature) <= 2048 is forced to zero; measured feature-MAC skip fraction: 57.81%

RTL debug counters: inference skips=22,199,950; learning-run inference skips=22,199,950; learning feature skips=5,603

Vivado top: cifar10_spikingformer_head_pynq_axi_stream_top part: xc7z020clg400-1 clock: 10 ns

Timing: WNS=0.336 ns WHS=0.100 ns timing_met=True

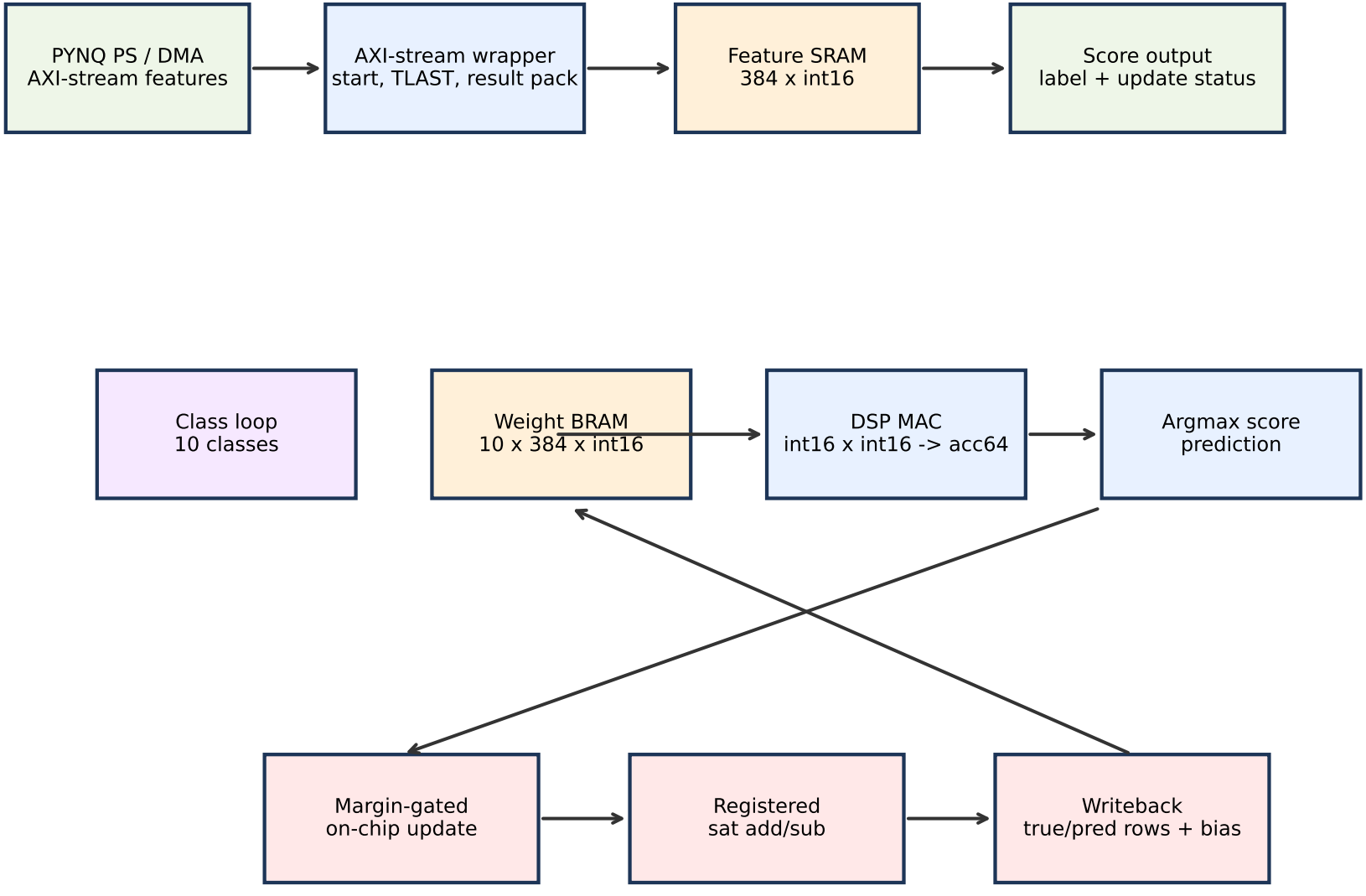
Utilization: LUT=1043 FF=988 BRAM36=6 DSP=1 LUTRAM=228

Power: total=0.123 W dynamic=0.018 W static=0.105 W

Mapping metrics file fields: bias_bits, checkpoint, confusion, feature_bits, feature_dim, feature_mac_skip_fraction, feature_scale, files, fixed_accuracy, float_accuracy

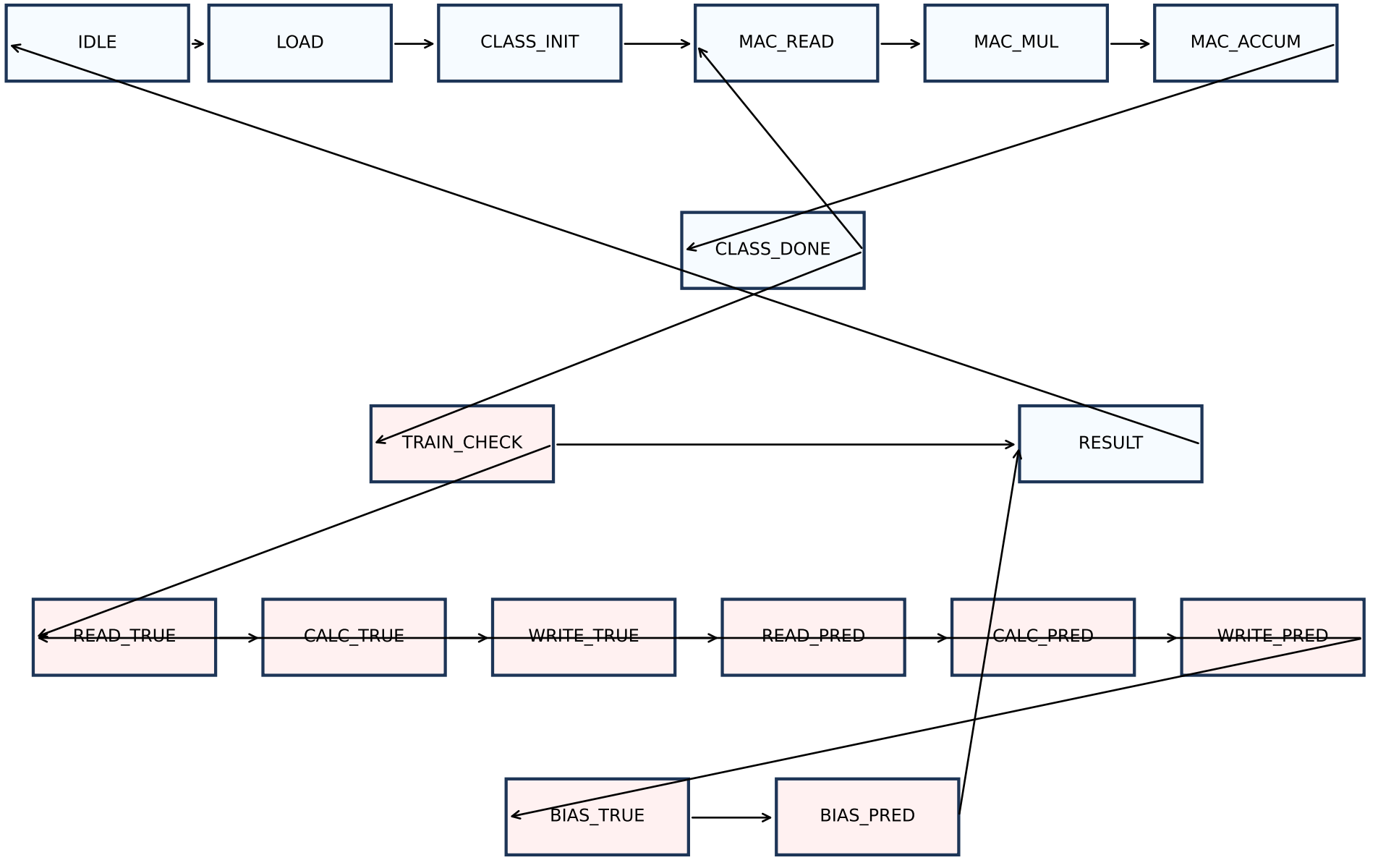
Important interpretation: the RTL accelerates the trained Spikingformer feature-head path, gates low-magnitude feature activity, and implements a margin-gated on-chip classifier update.

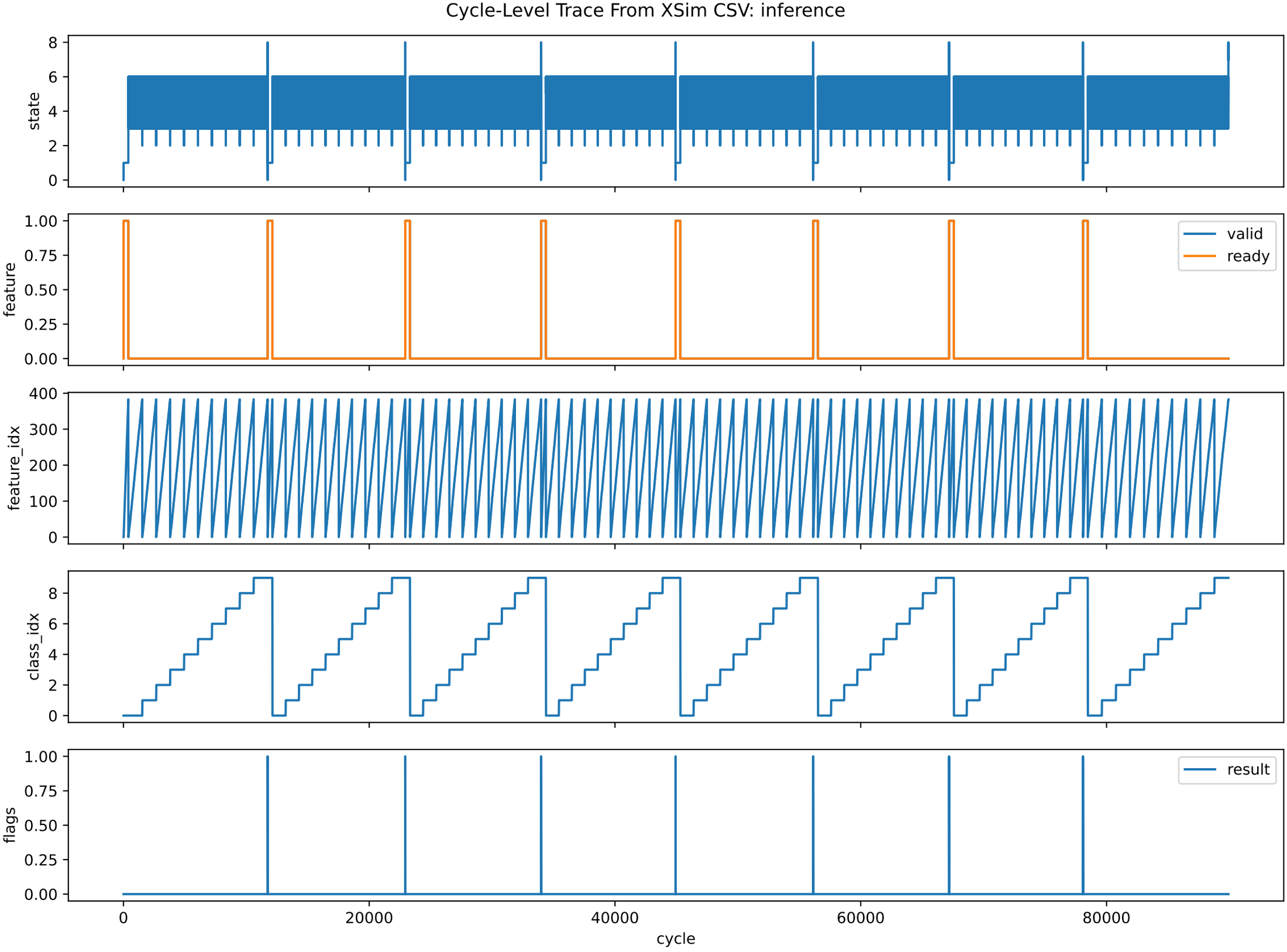
NeuraEdge CIFAR-10 Spikingformer Head Accelerator Dataflow

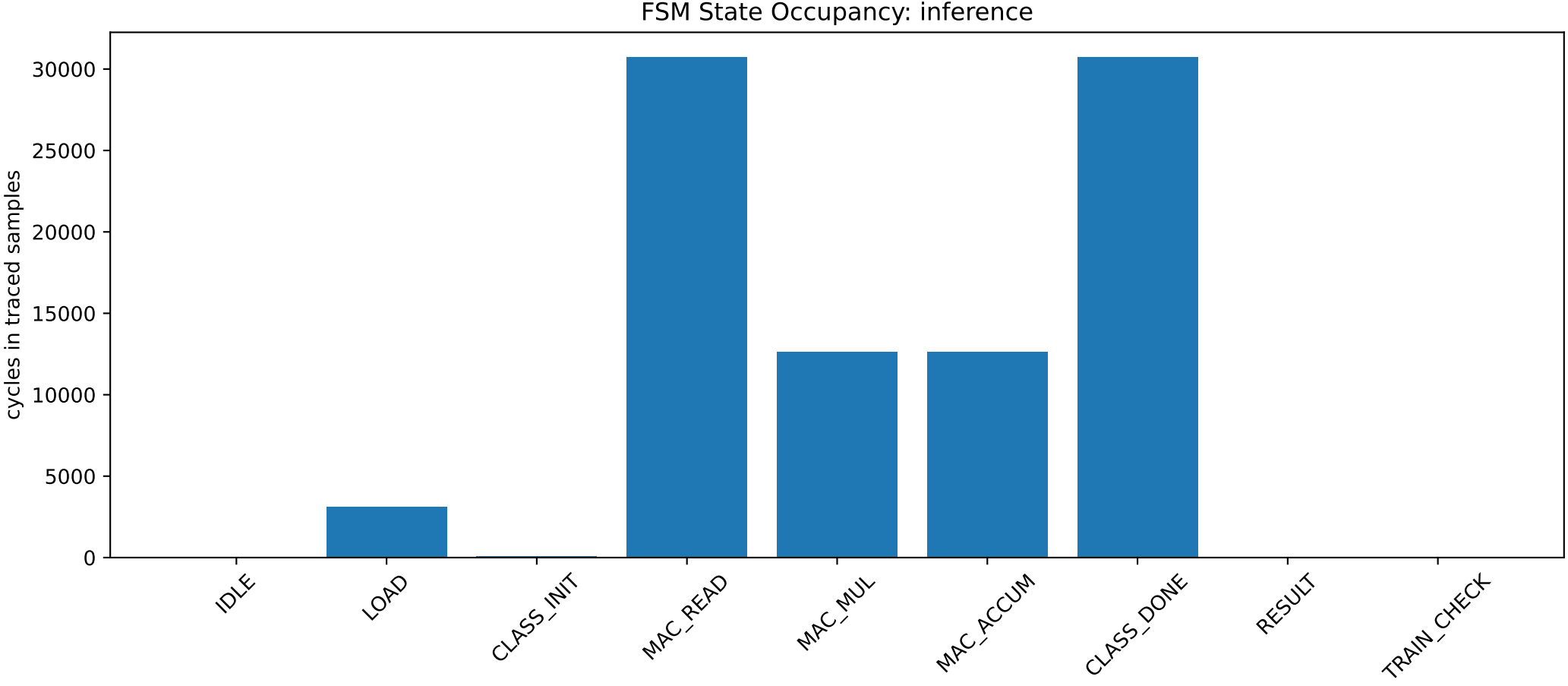


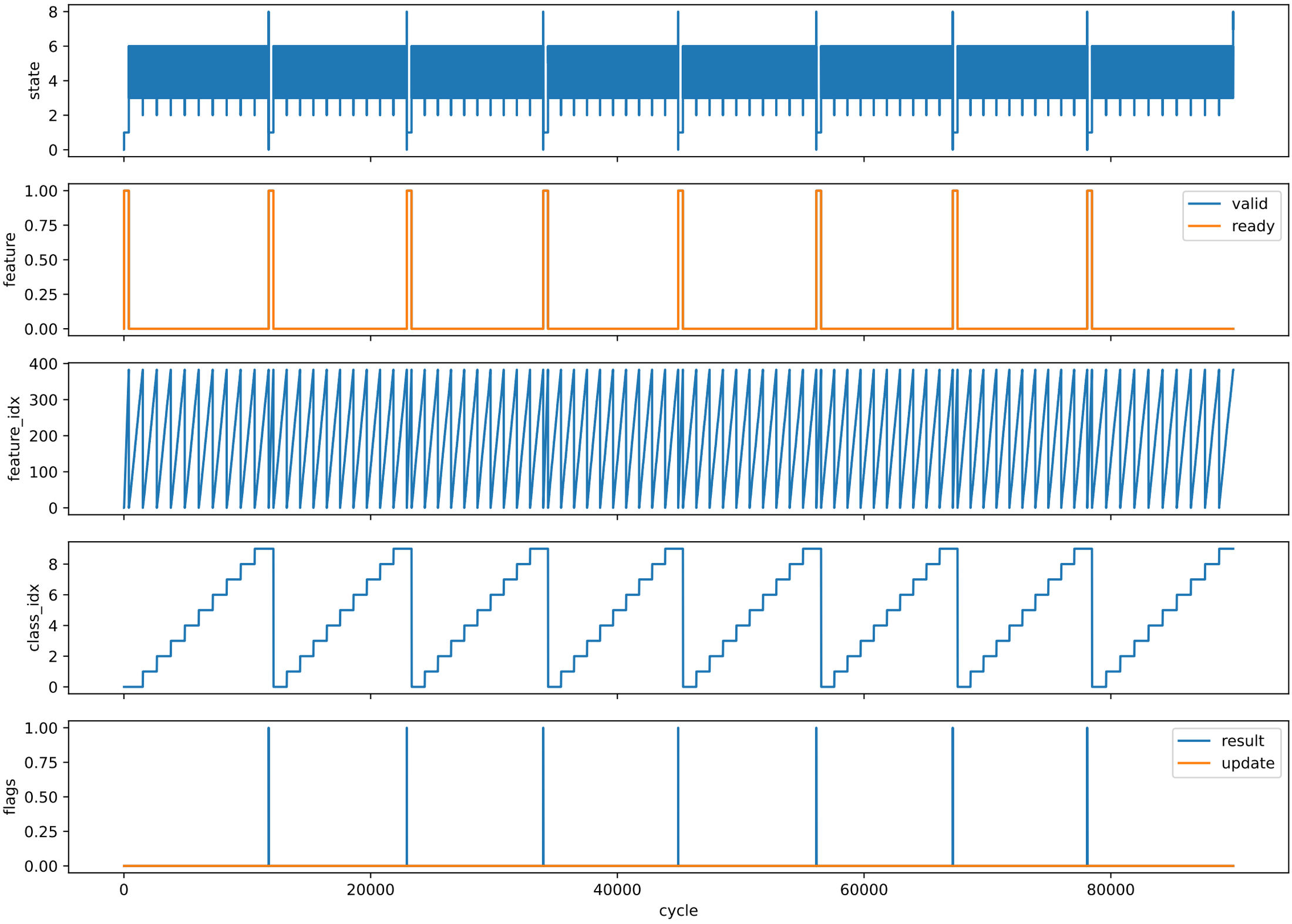
All numbers in this report come from the generated RTL memories, XSim CSV traces, and Vivado reports in outputs_spikingformer_head_accelerator.

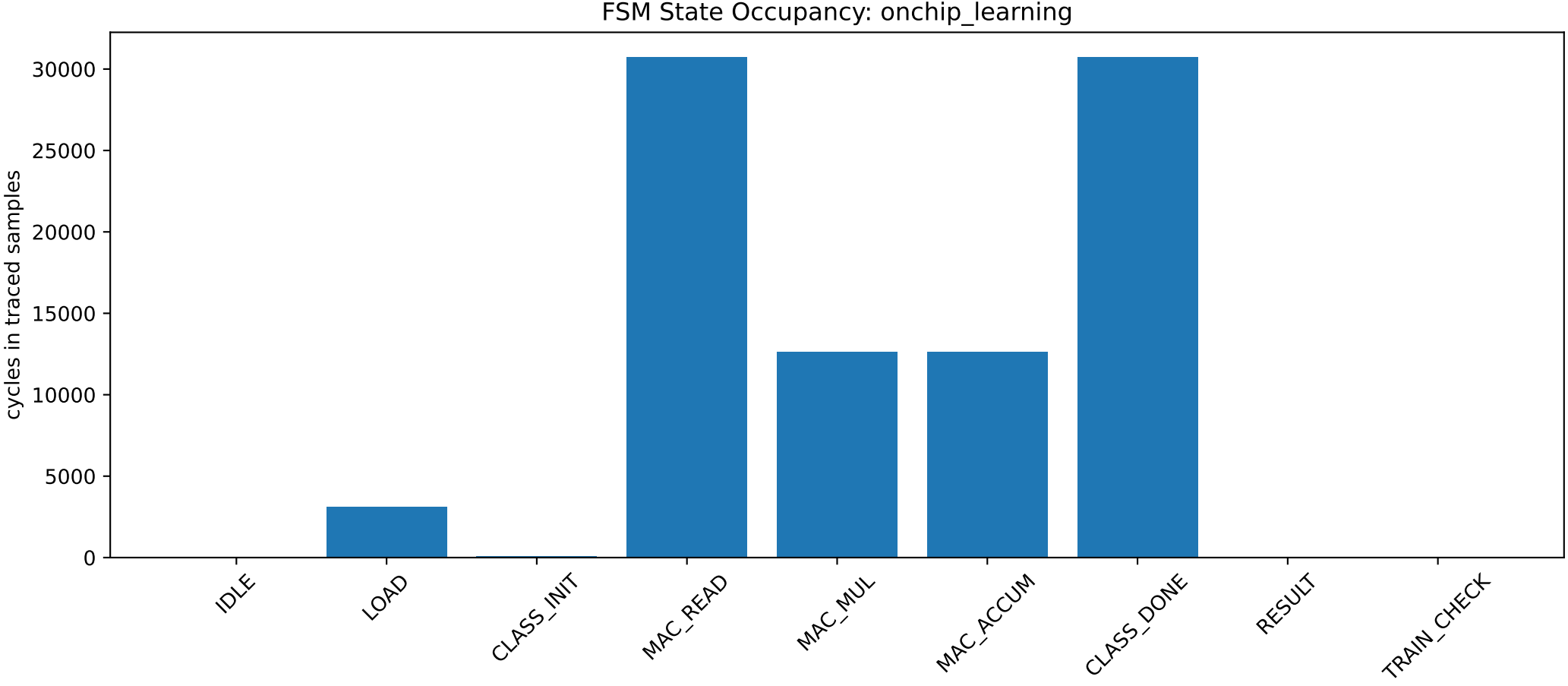
RTL State Machine Including On-Chip Learning

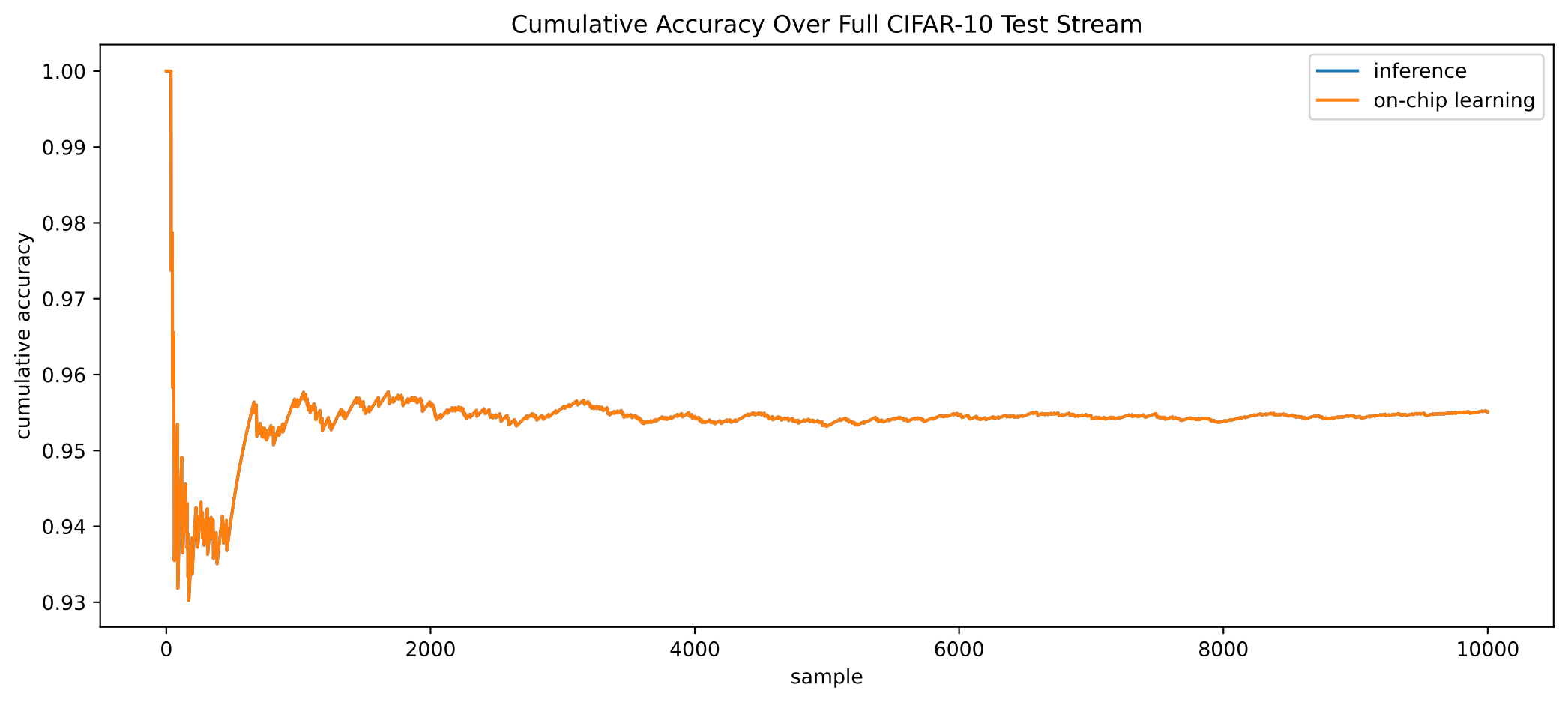


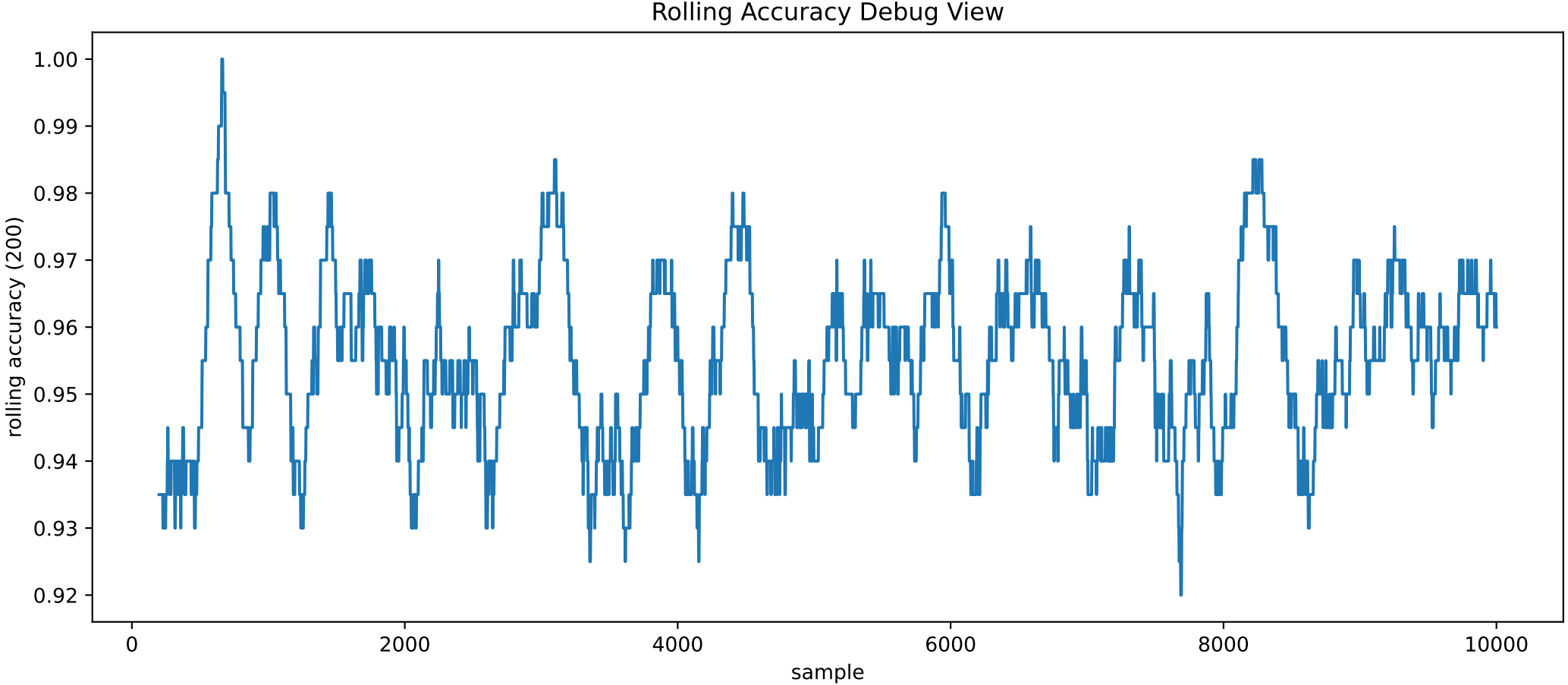

Cycle-Level Trace From XSim CSV: inference
state
feature
valid
ready
feature_idx
class_idx
flags
result
cycle
0
20000
40000
60000
80000

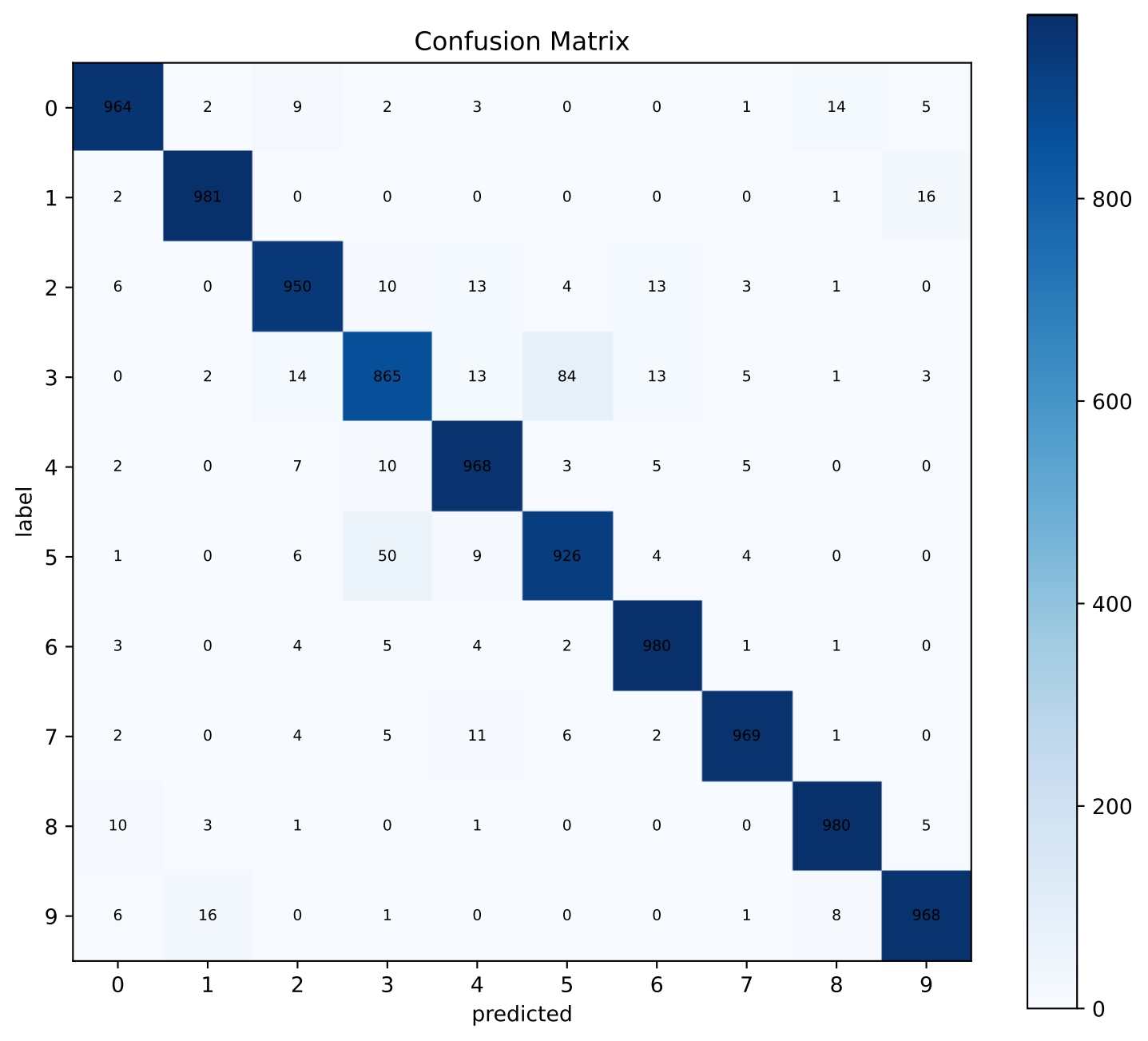

FSM State Occupancy: inference
cycles in traced samples
0
5000
10000
15000
20000
25000
30000
IDLE
LOAD
CLASS_INIT
MAC_READ
MAC_MUL
MAC_ACCUM
CLASS_DONE
RESULT
TRAIN_CHECK

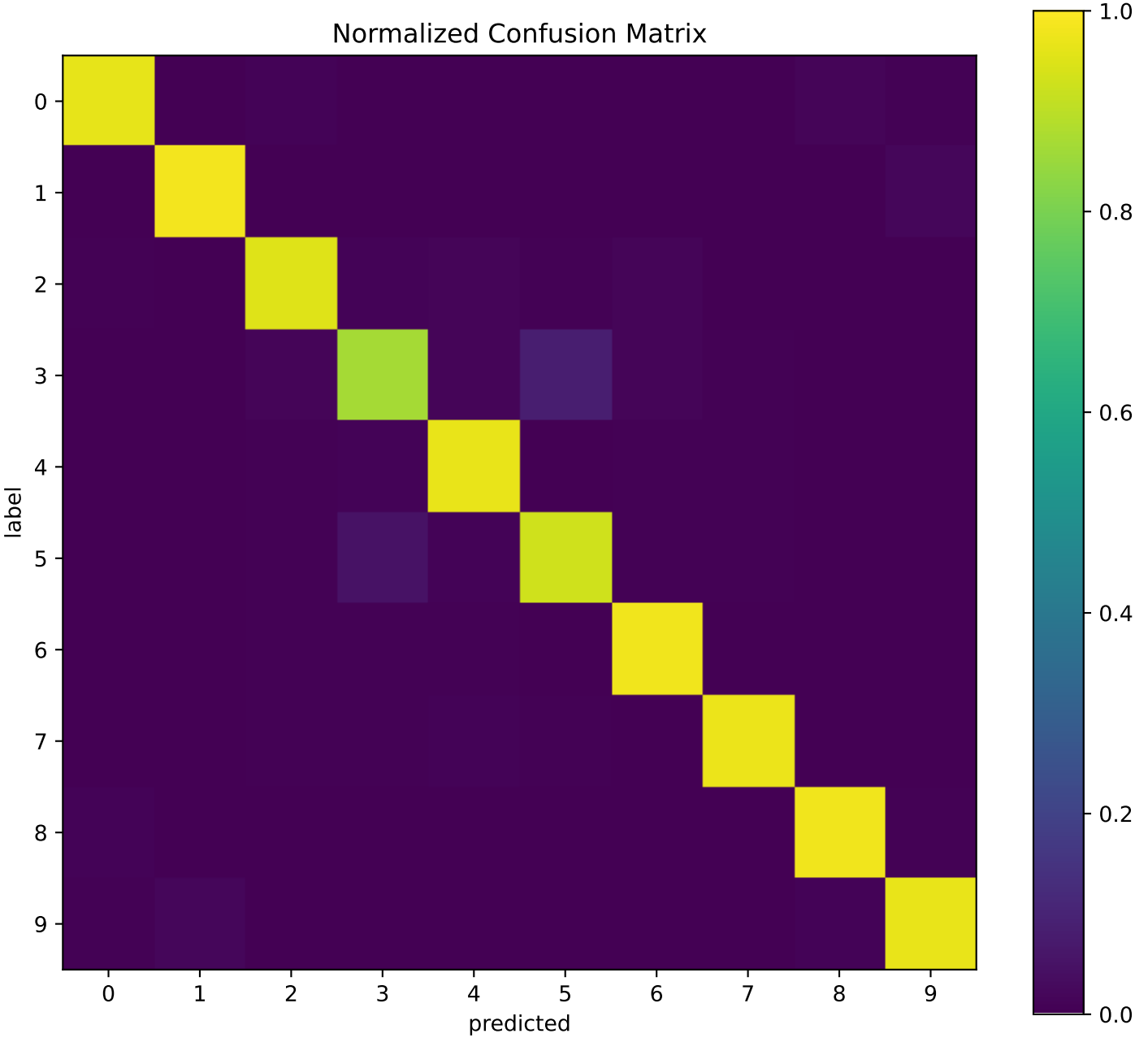

Cycle-Level Trace From XSim CSV: onchip_learning
state
feature
valid
ready
feature_idx
class_idx
flags
result
update
cycle
0
20000
40000
60000
80000

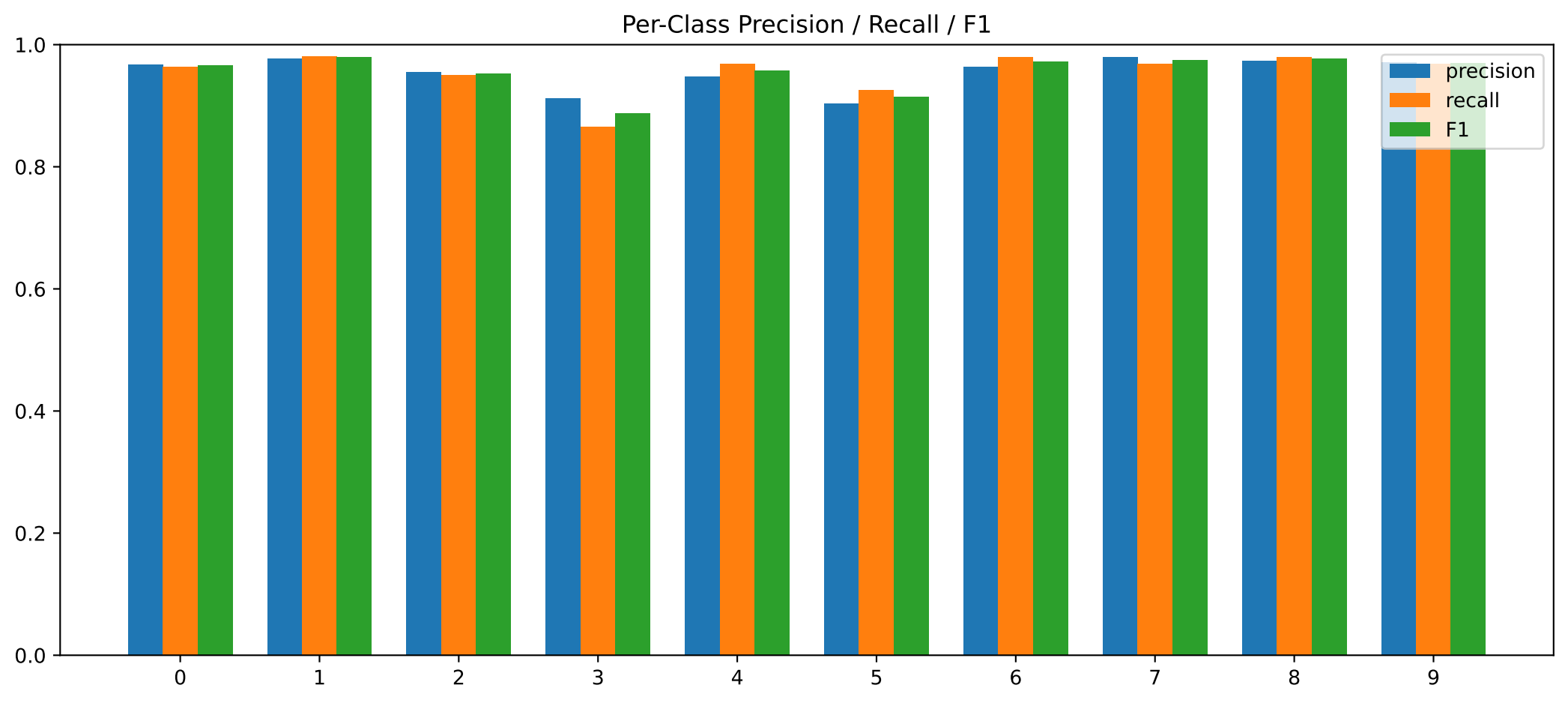

FSM State Occupancy: onchip_learning
cycles in traced samples
0
5000
10000
15000
20000
25000
30000
IDLE
LOAD
CLASS_INIT
MAC_READ
MAC_MUL
MAC_ACCUM
CLASS_DONE
RESULT
TRAIN_CHECK

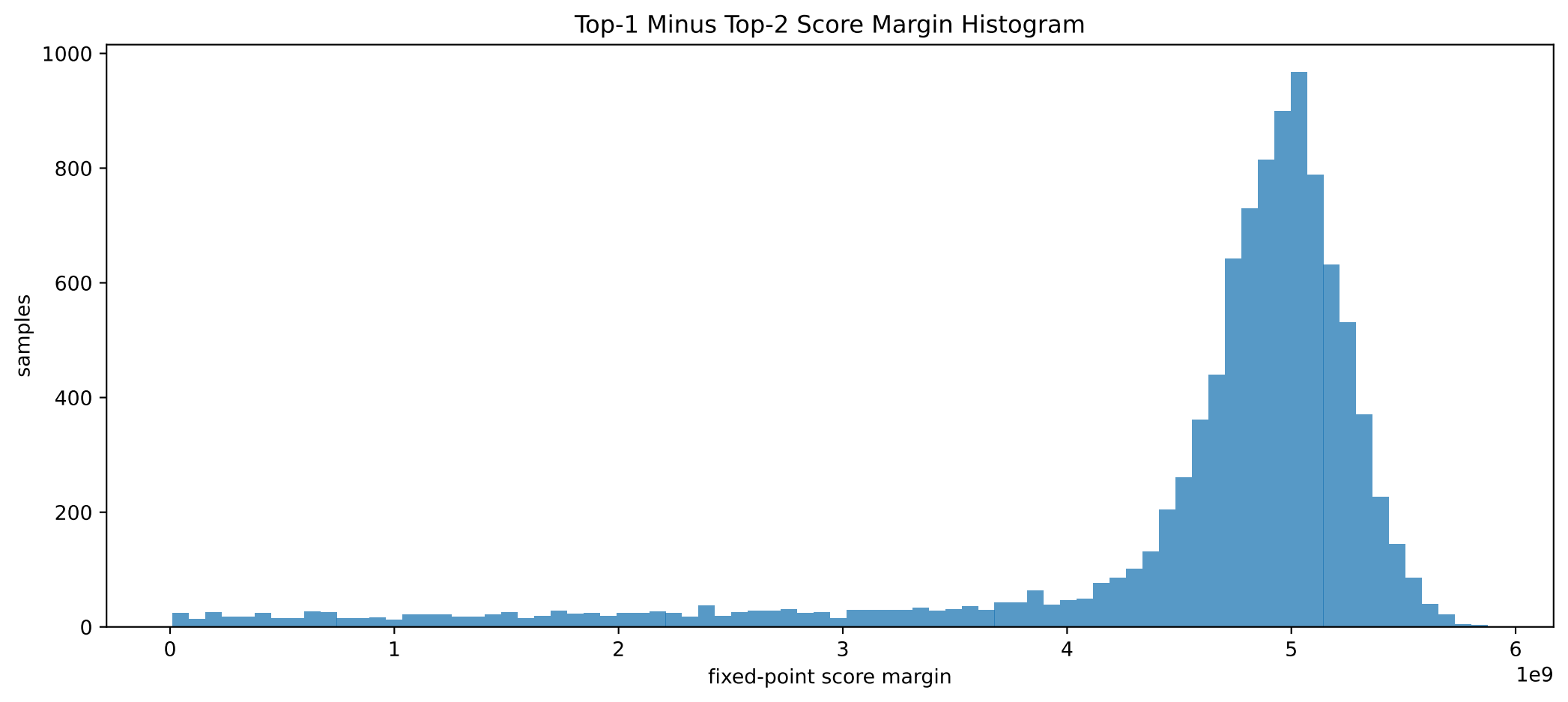

Cumulative Accuracy Over Full CIFAR-10 Test Stream
inference
on-chip learning
cumulative accuracy
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0
2000
4000
6000
8000
10000
sample

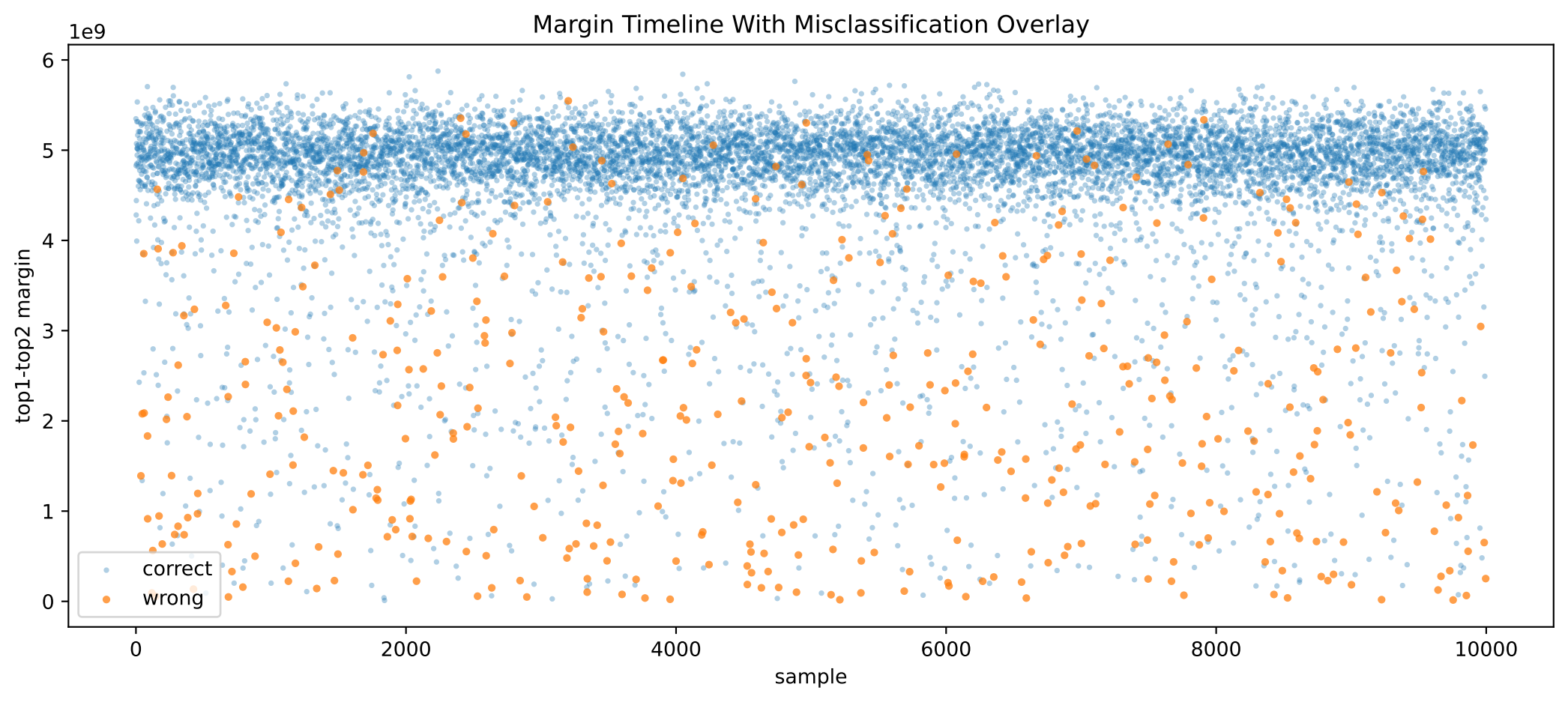

Rolling Accuracy Debug View
rolling accuracy (200)
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92
0
2000
4000
6000
8000
10000
sample

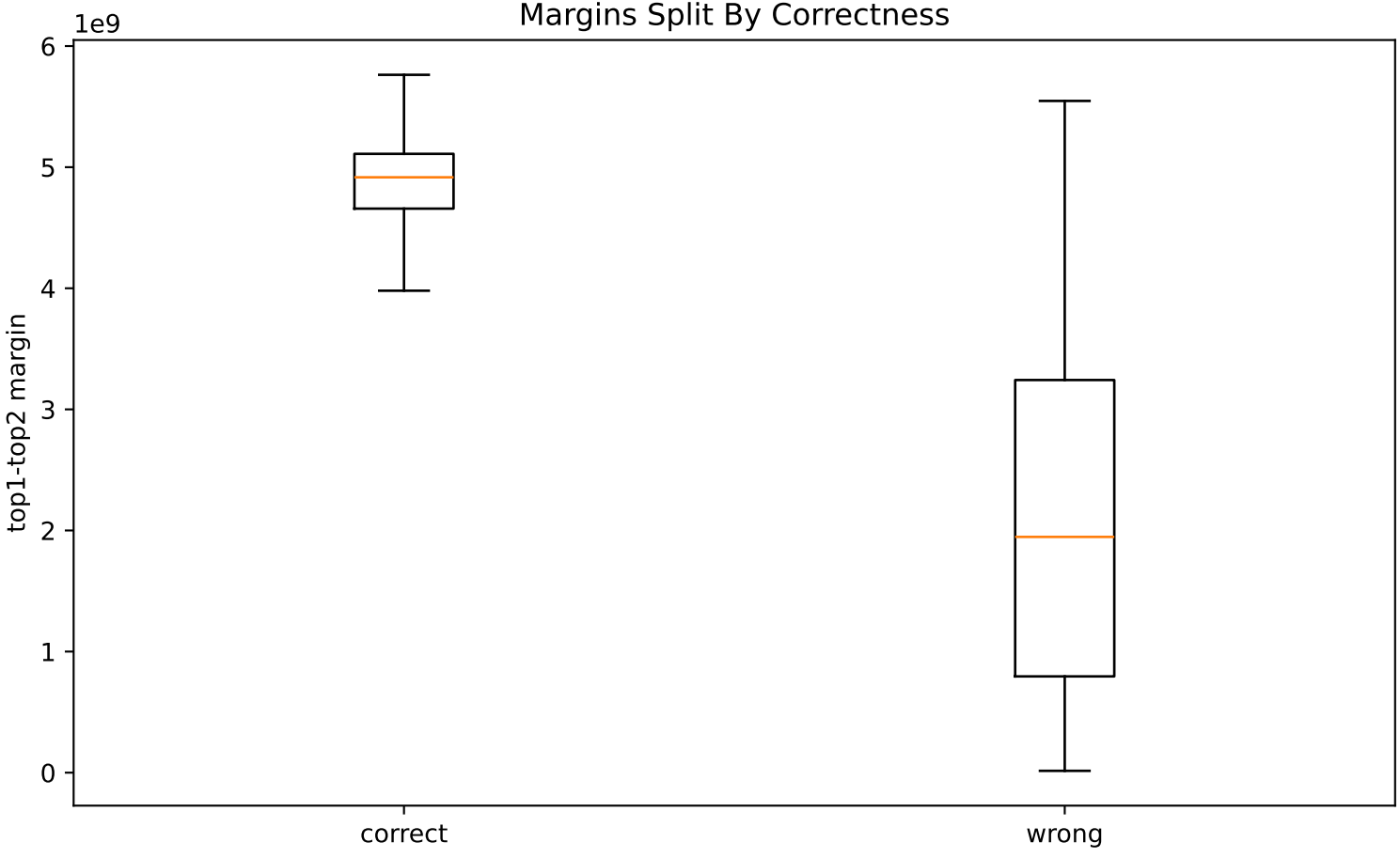

Confusion Matrix
label
predicted
0 964 2 9 2 3 0 0 1 14 5
1 2 981 0 0 0 0 0 0 1 16
2 6 0 950 10 13 4 13 3 1 0
3 0 2 14 865 13 84 13 5 1 3
4 2 0 7 10 968 3 5 5 0 0
5 1 0 6 50 9 926 4 4 0 0
6 3 0 4 5 4 2 980 1 1 0
7 2 0 4 5 11 6 2 969 1 0
8 10 3 1 0 1 0 0 0 980 5
9 6 16 0 1 0 0 0 1 8 968
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

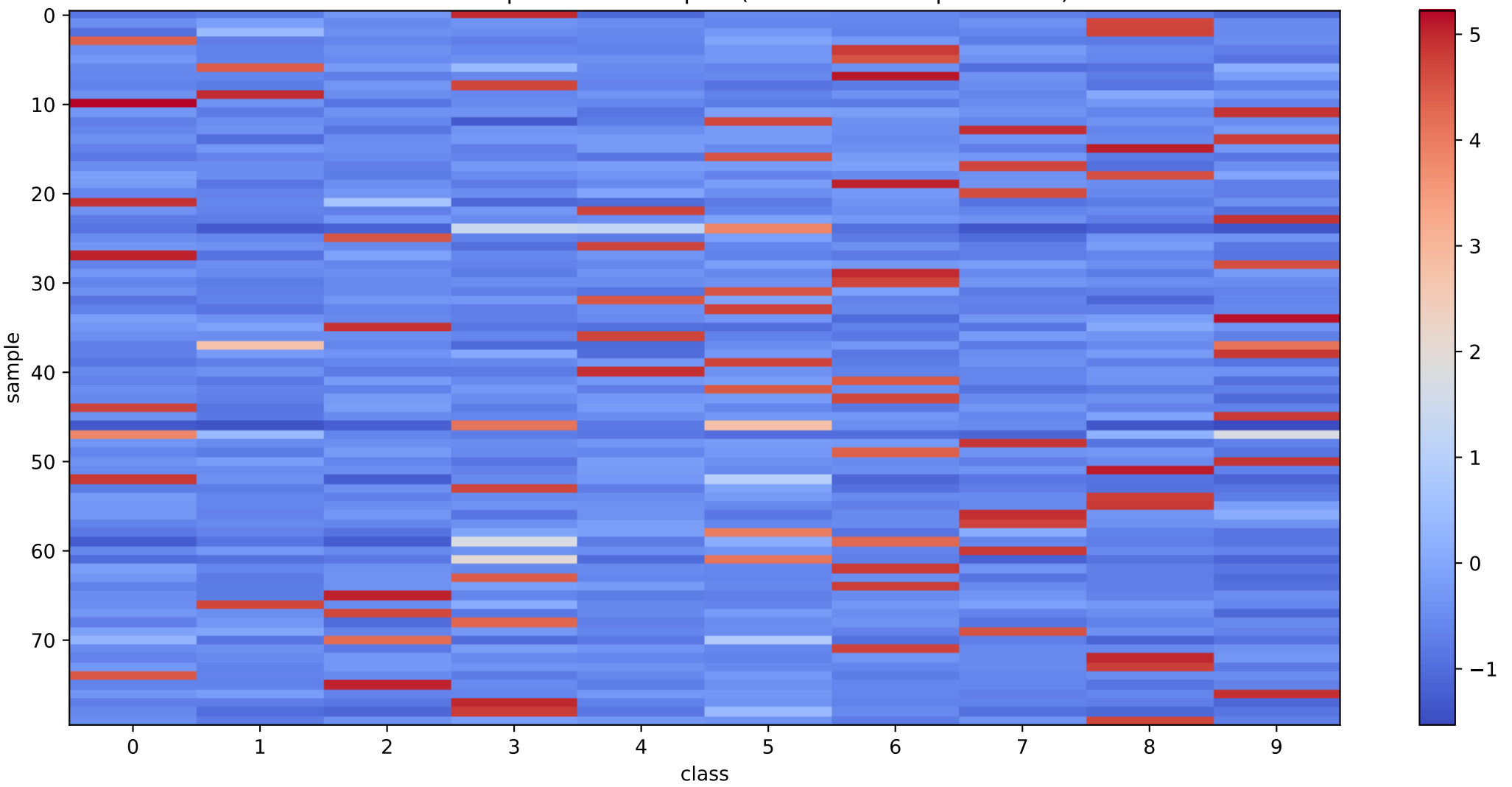

Normalized Confusion Matrix
label
predicted
0
1
2
3
4
5
6
7
8
9
1.0
0.8
0.6
0.4
0.2
0.0

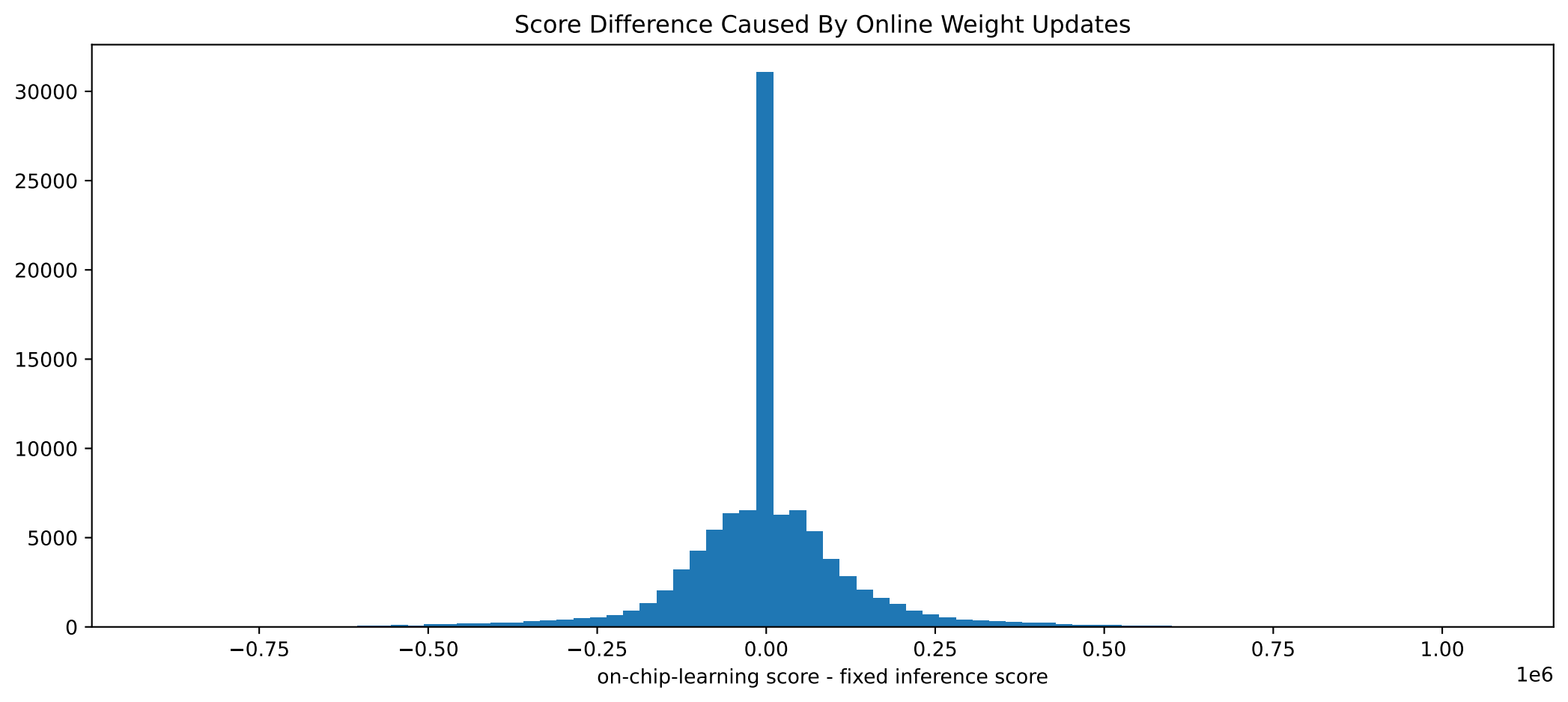

Per-Class Precision / Recall / F1
precision
recall
F1
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
6
7
8
9

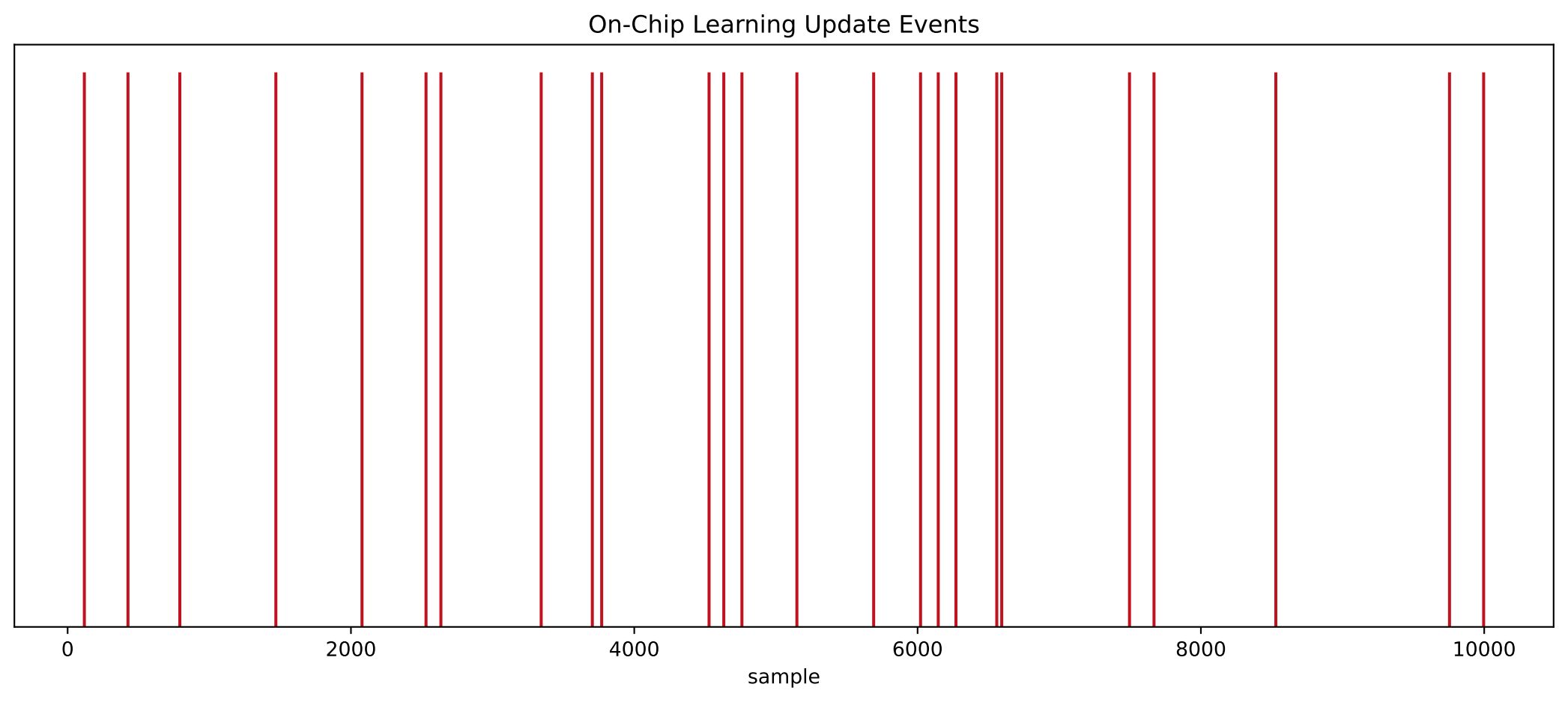

Top-1 Minus Top-2 Score Margin Histogram
samples
0
200
400
600
800
1000
0
1
2
3
4
5
6
1e9
fixed-point score margin

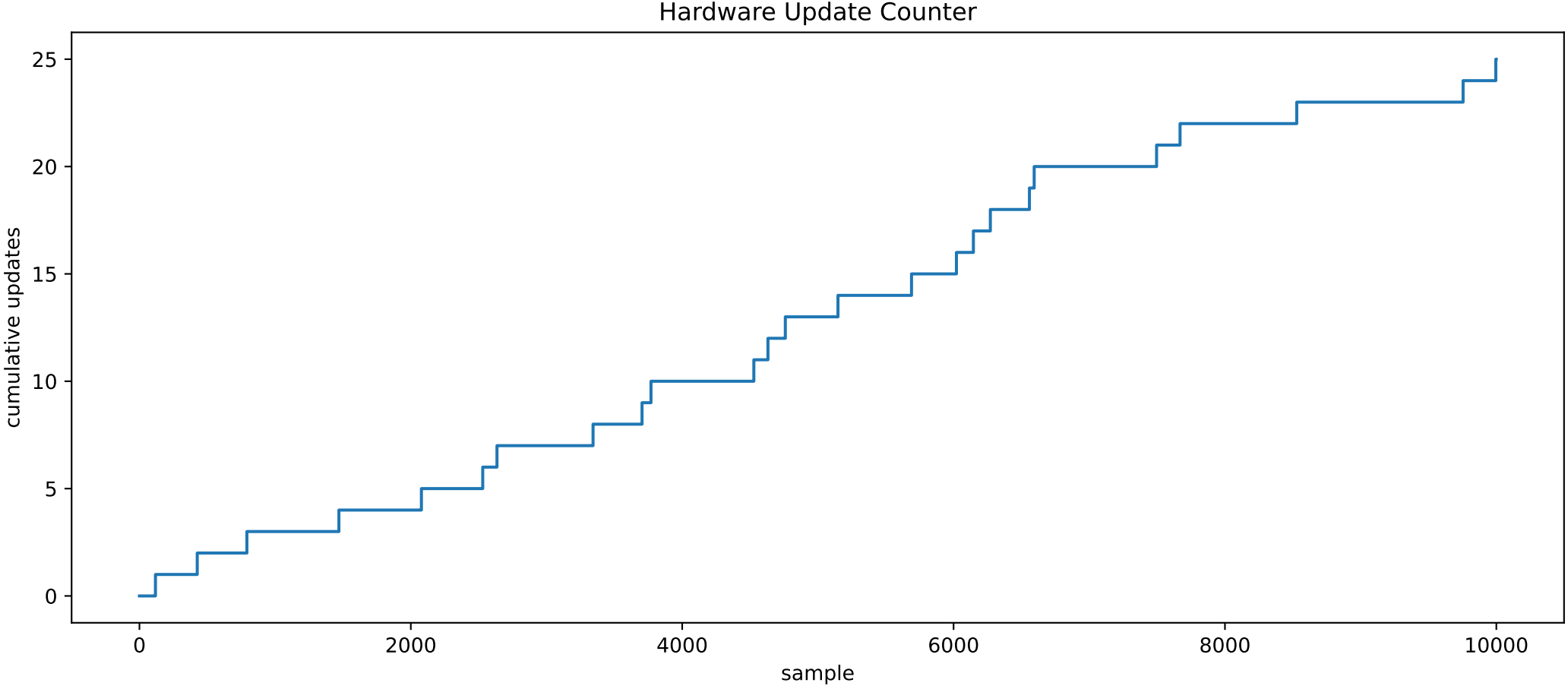

Margin Timeline With Misclassification Overlay
1e9
top1-top2 margin
sample
correct
wrong
0
1
2
3
4
5
6
0
2000
4000
6000
8000
10000

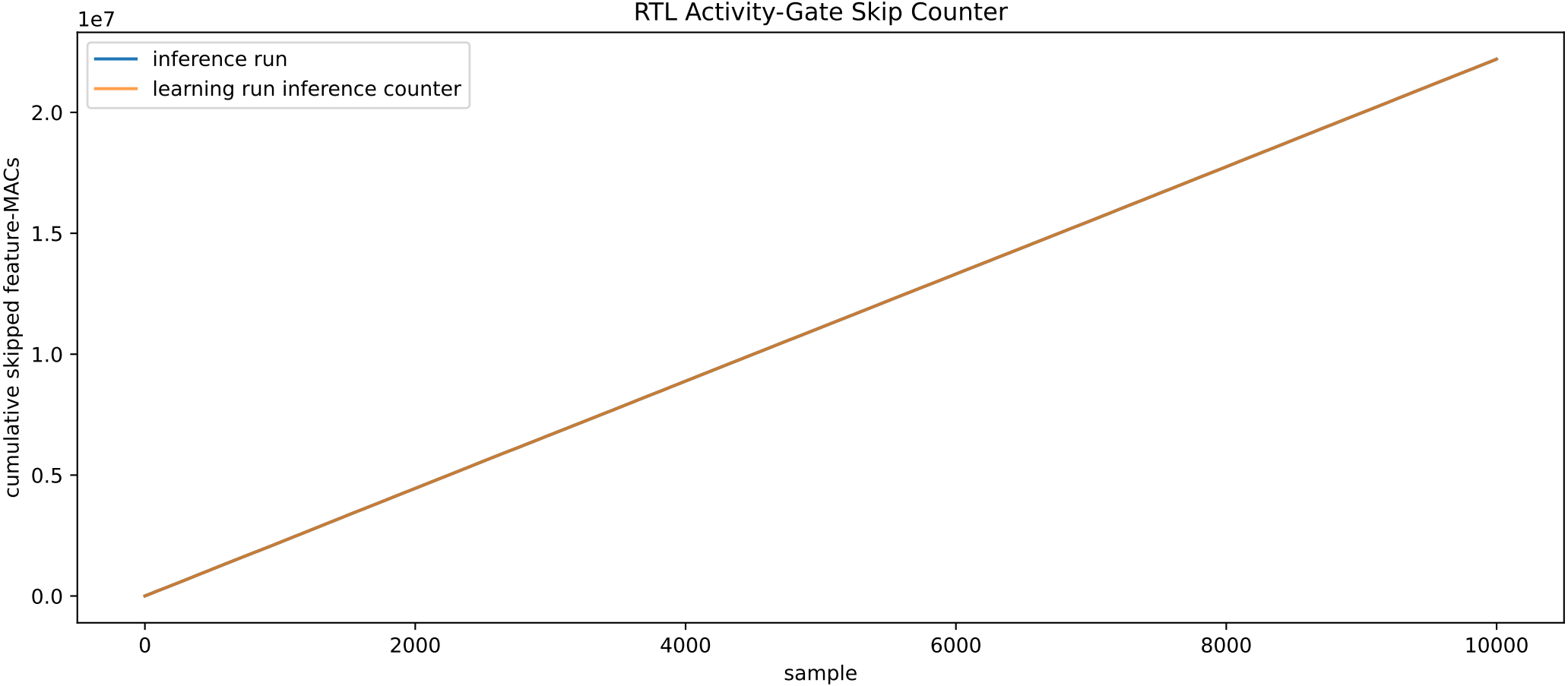

Margins Split By Correctness
1e9
6
5
4
3
2
1
0
top1-top2 margin
correct
wrong

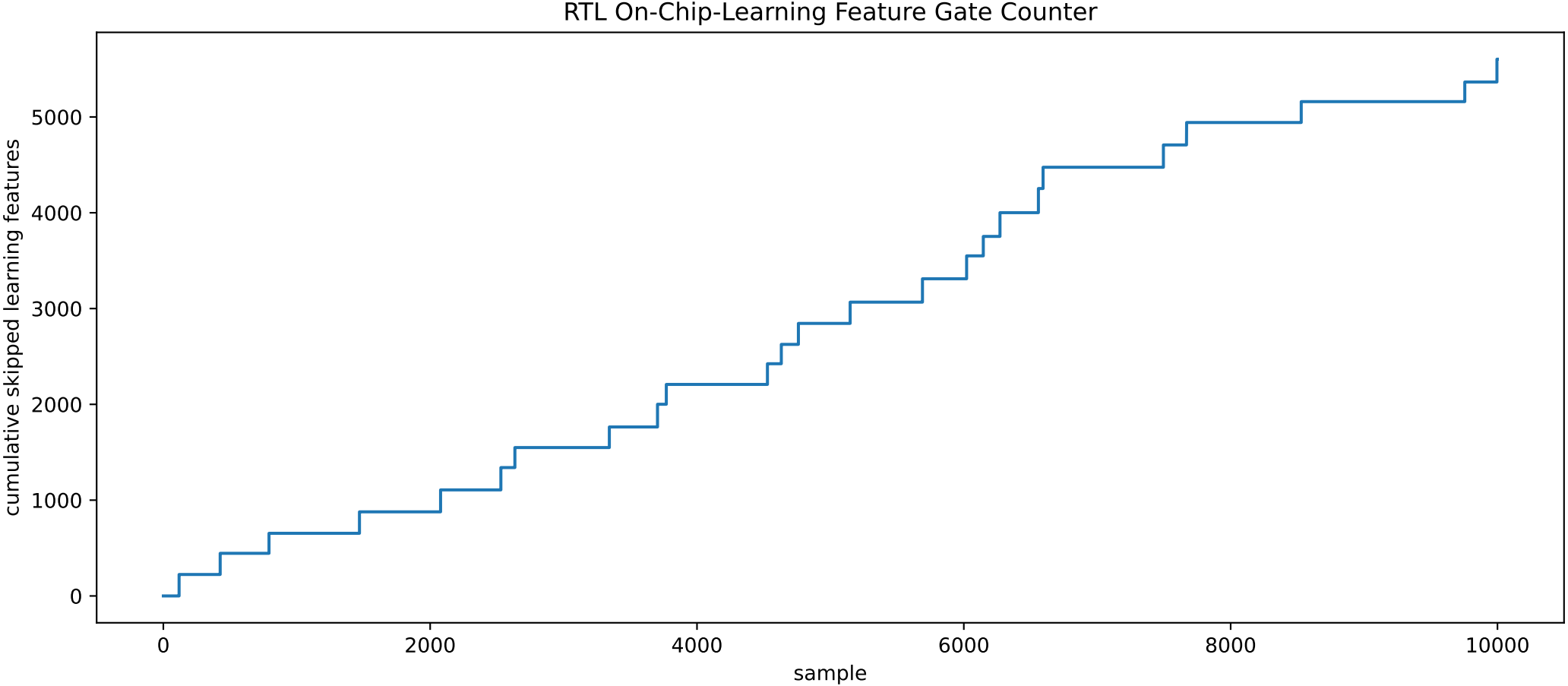

Score Heatmap: First 80 Samples (billions of fixed-point units)
sample
0
10
20
30
40
50
60
70
class
0
1
2
3
4
5
6
7
8
9
5
4
3
2
1
0
−1

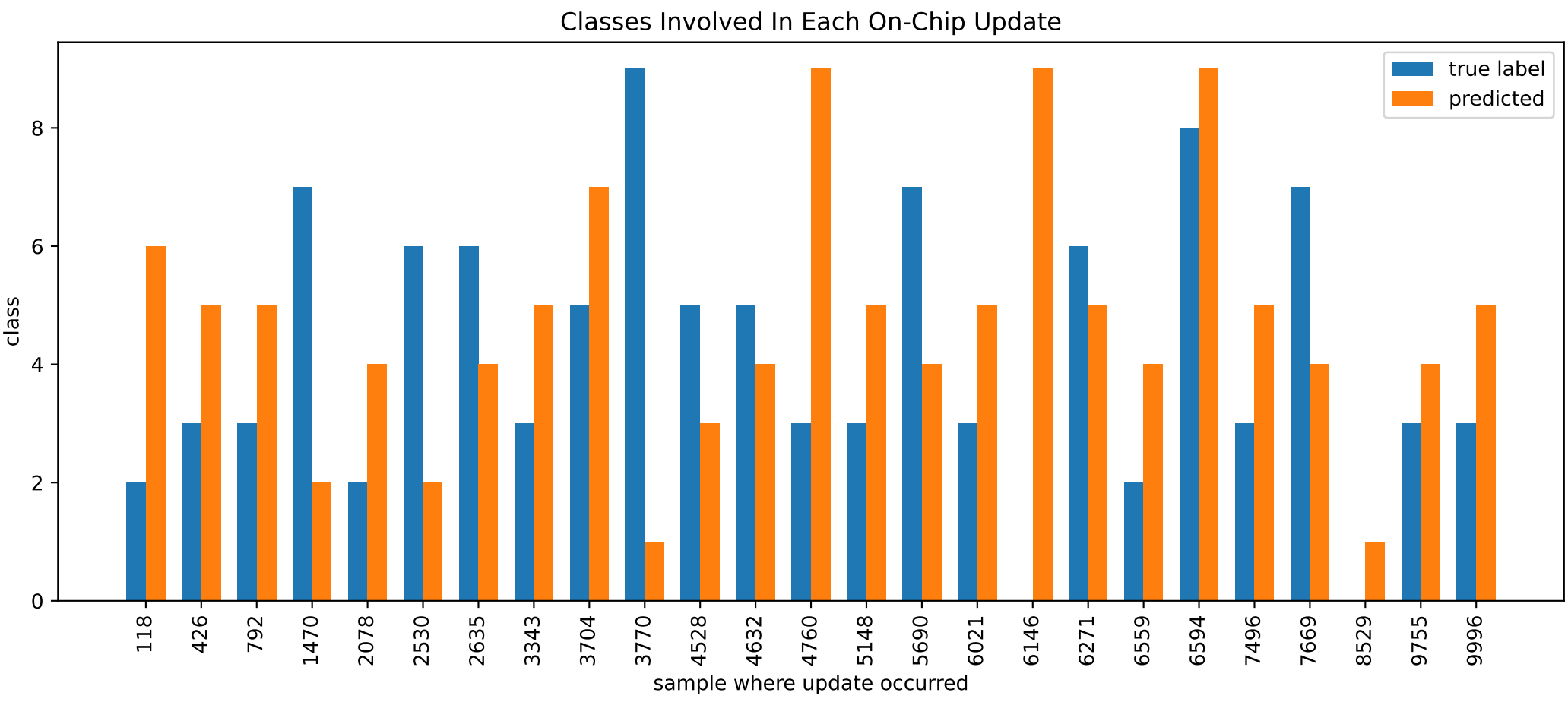

Score Difference Caused By Online Weight Updates
30000
25000
20000
15000
10000
5000
0
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
on-chip-learning score - fixed inference score
1e6

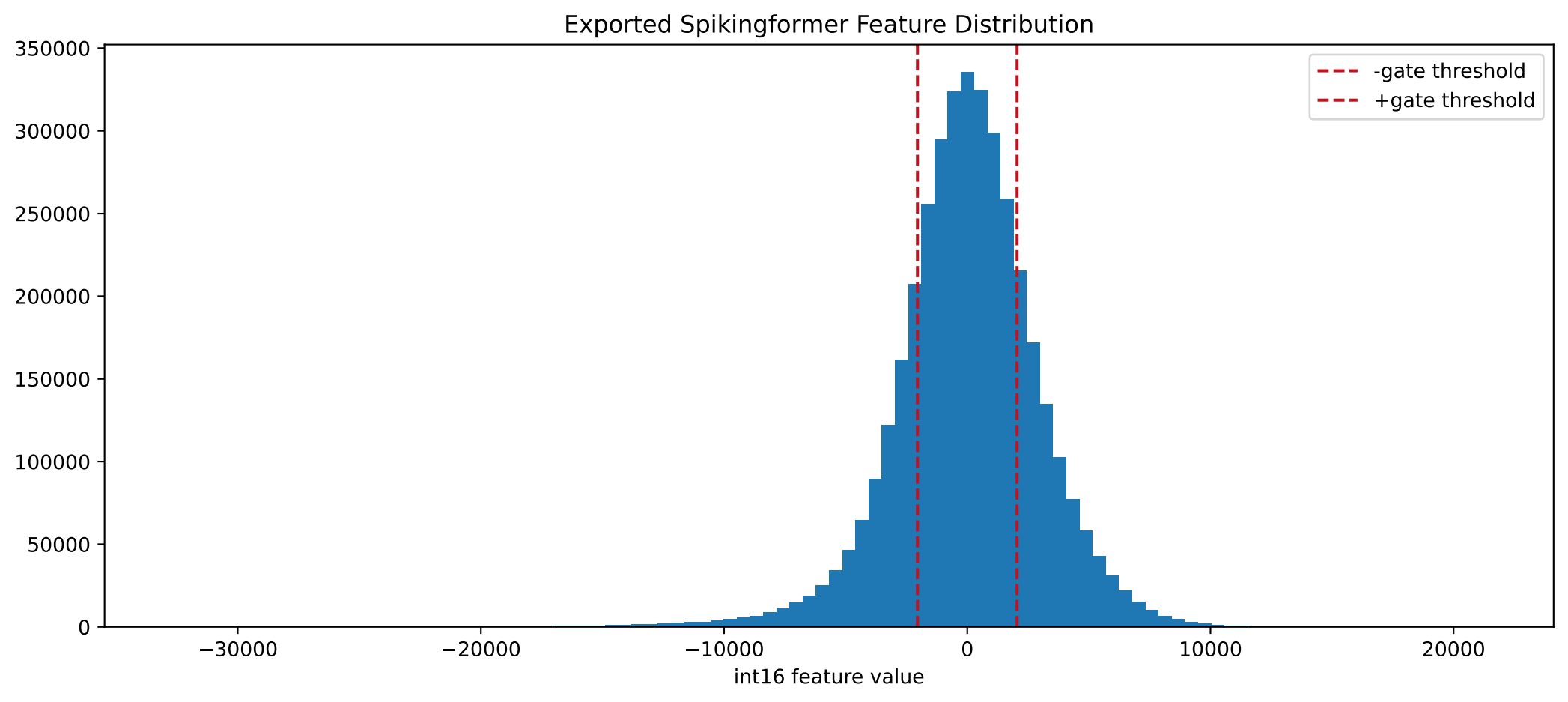

On-Chip Learning Update Events
0
2000
4000
6000
8000
10000
sample

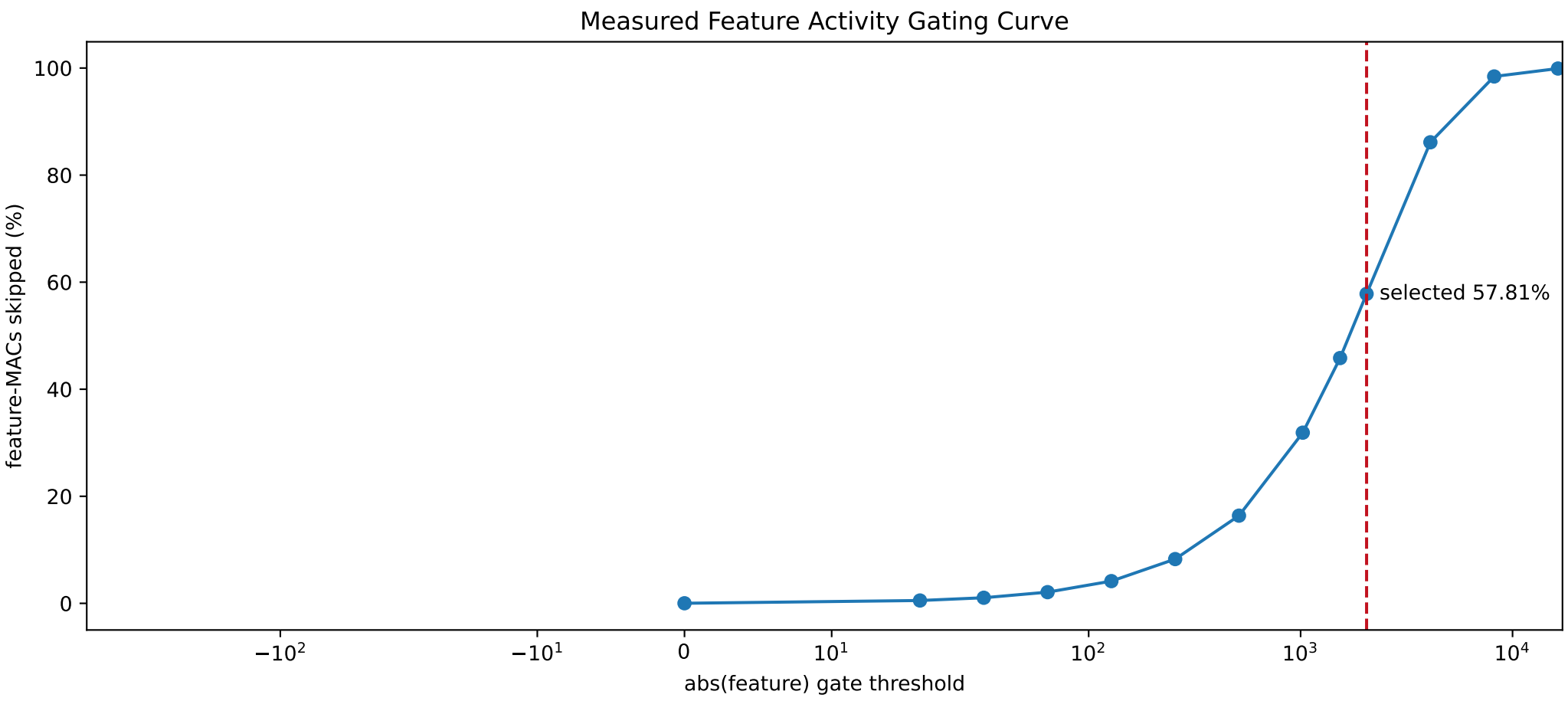

Hardware Update Counter
cumulative updates
sample
0
5
10
15
20
25
0
2000
4000
6000
8000
10000

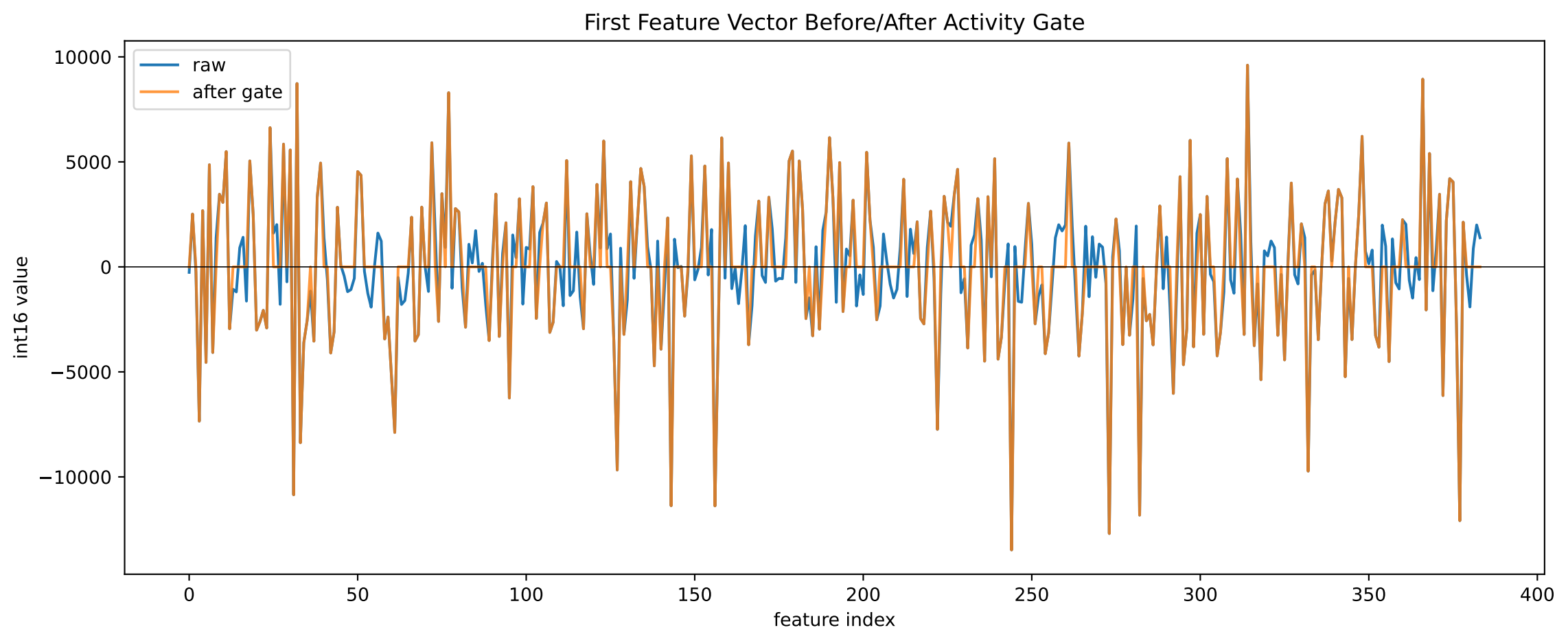

RTL Activity-Gate Skip Counter
1e7
inference run
learning run inference counter
cumulative skipped feature-MACs
2.0
1.5
1.0
0.5
0.0
0
2000
4000
6000
8000
10000
sample

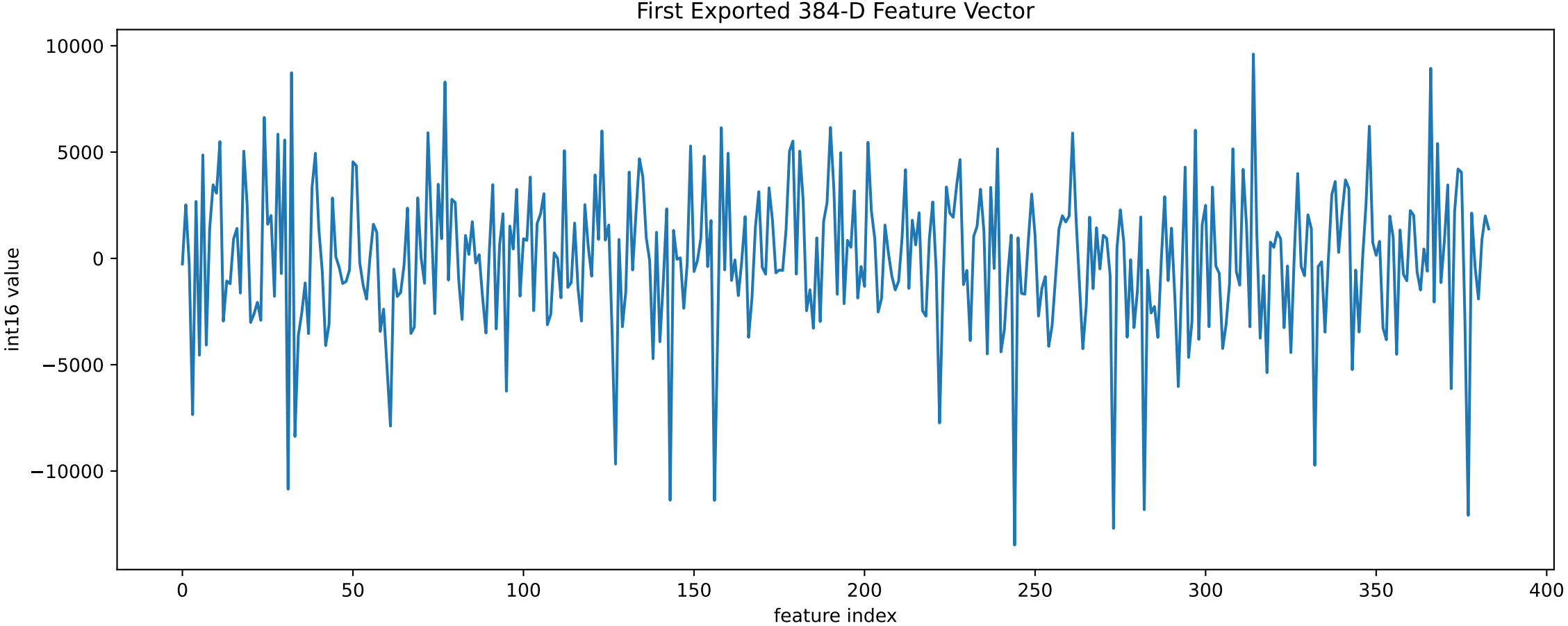

RTL On-Chip-Learning Feature Gate Counter
cumulative skipped learning features
0
1000
2000
3000
4000
5000
0
2000
4000
6000
8000
10000
sample

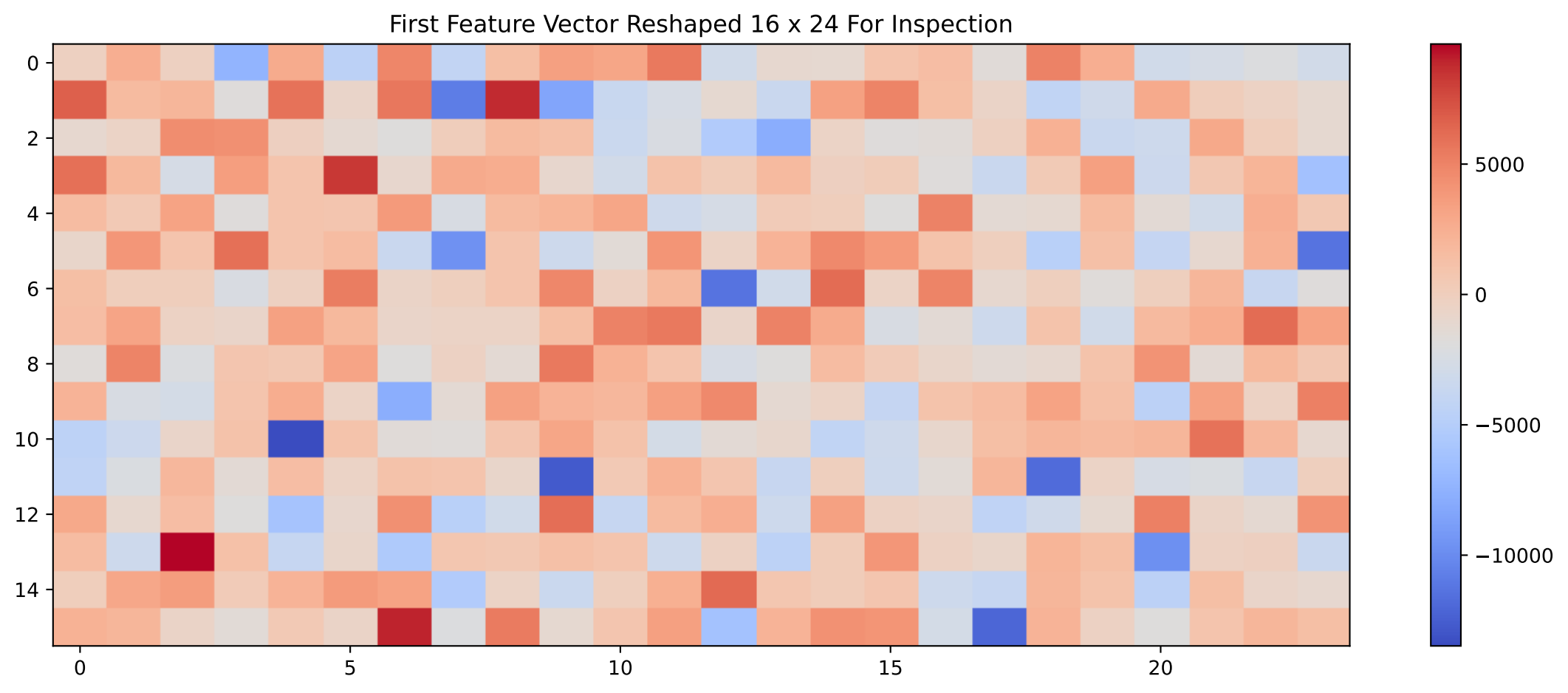

Classes Involved In Each On-Chip Update
true label
predicted
class
0
2
4
6
8
118
426
792
1470
2078
2530
2635
3343
3704
3770
4528
4632
4760
5148
5690
6021
6146
6271
6559
6594
7496
7669
8529
9755
9996
sample where update occurred

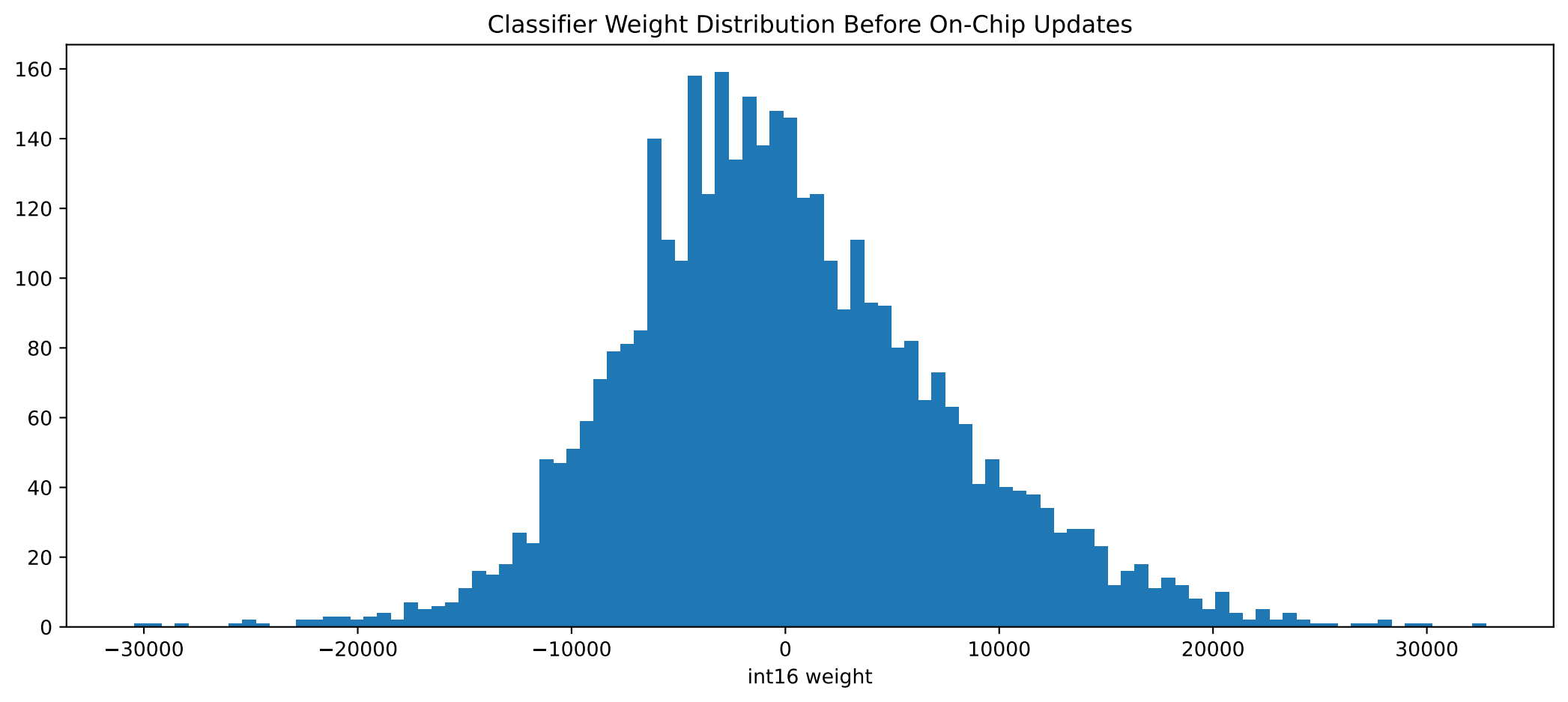

Exported Spikingformer Feature Distribution
-gate threshold
+gate threshold
350000
300000
250000
200000
150000
100000
50000
0
−30000
−20000
−10000
0
10000
20000
int16 feature value

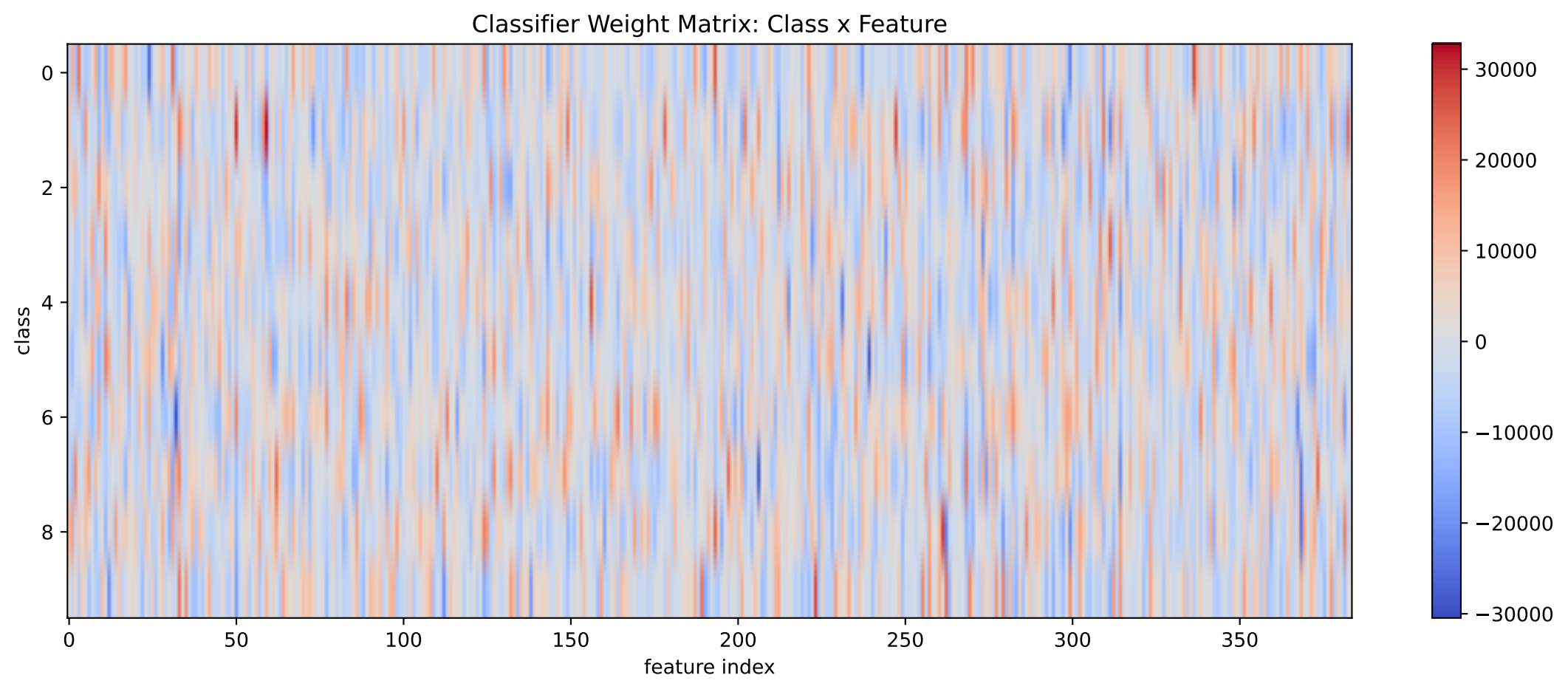

Measured Feature Activity Gating Curve
feature-MACs skipped (%)
100
80
60
40
20
0
selected 57.81%
-10^2
-10^1
0
10^1
10^2
10^3
10^4
abs(feature) gate threshold

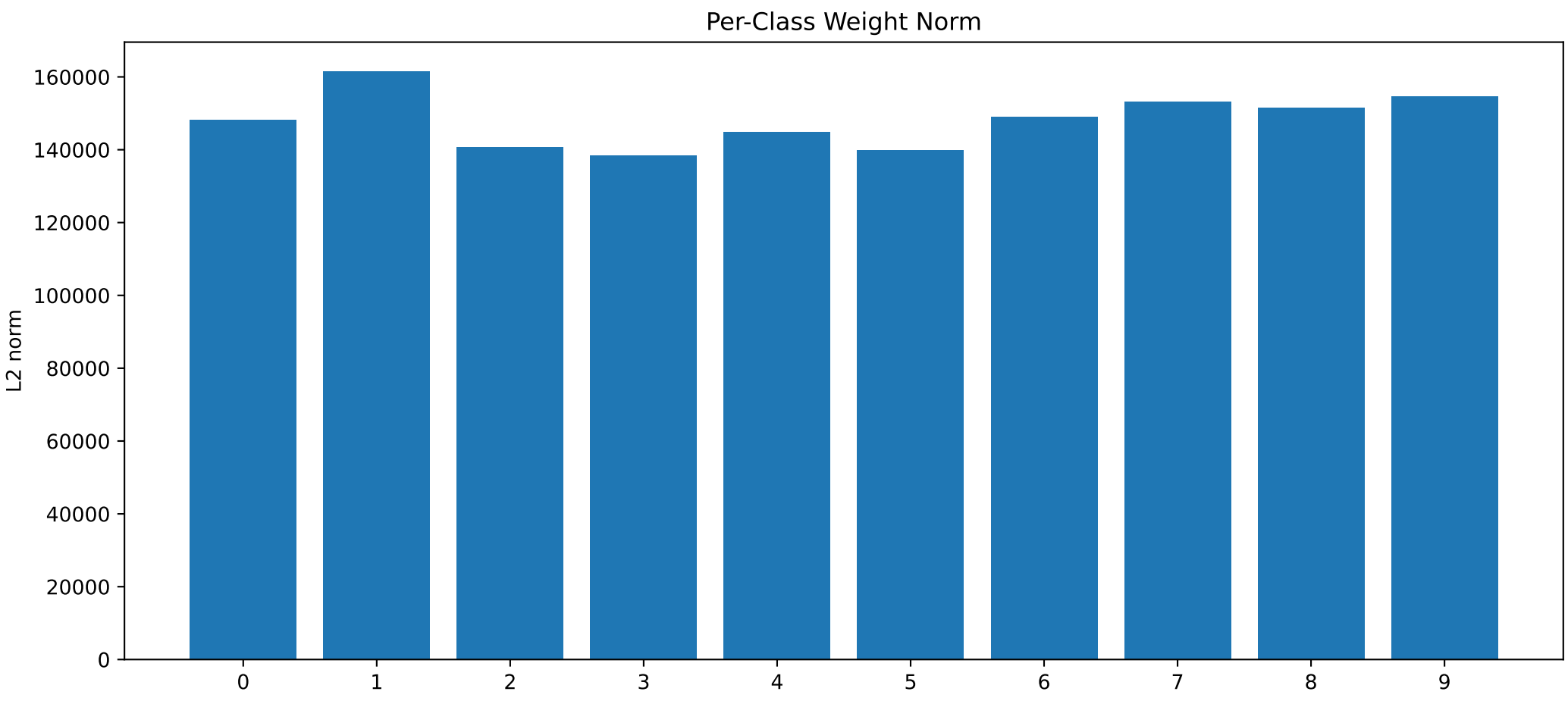

First Feature Vector Before/After Activity Gate
raw
after gate
10000
5000
0
−5000
−10000
int16 value
0
50
100
150
200
250
300
350
400
feature index

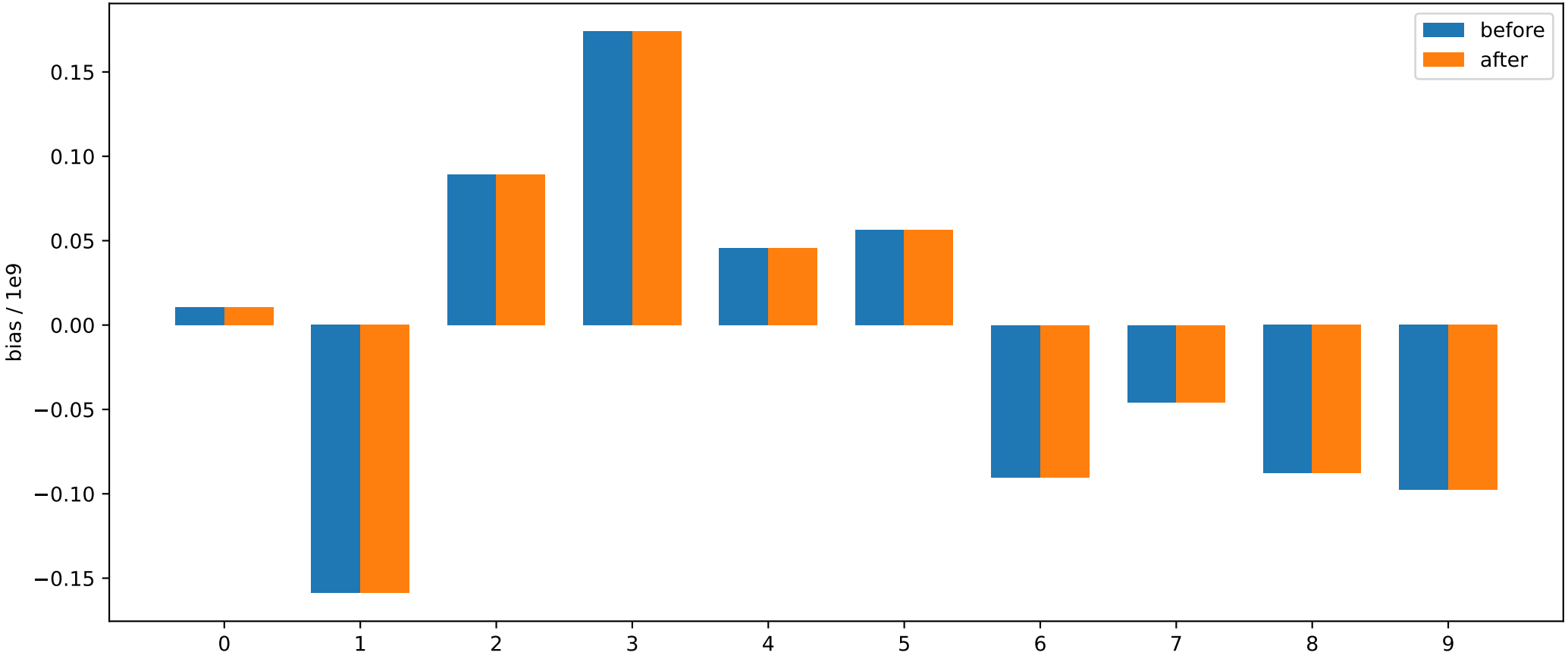

First Exported 384-D Feature Vector
int16 value
feature index
10000
5000
0
−5000
−10000
0
50
100
150
200
250
300
350
400

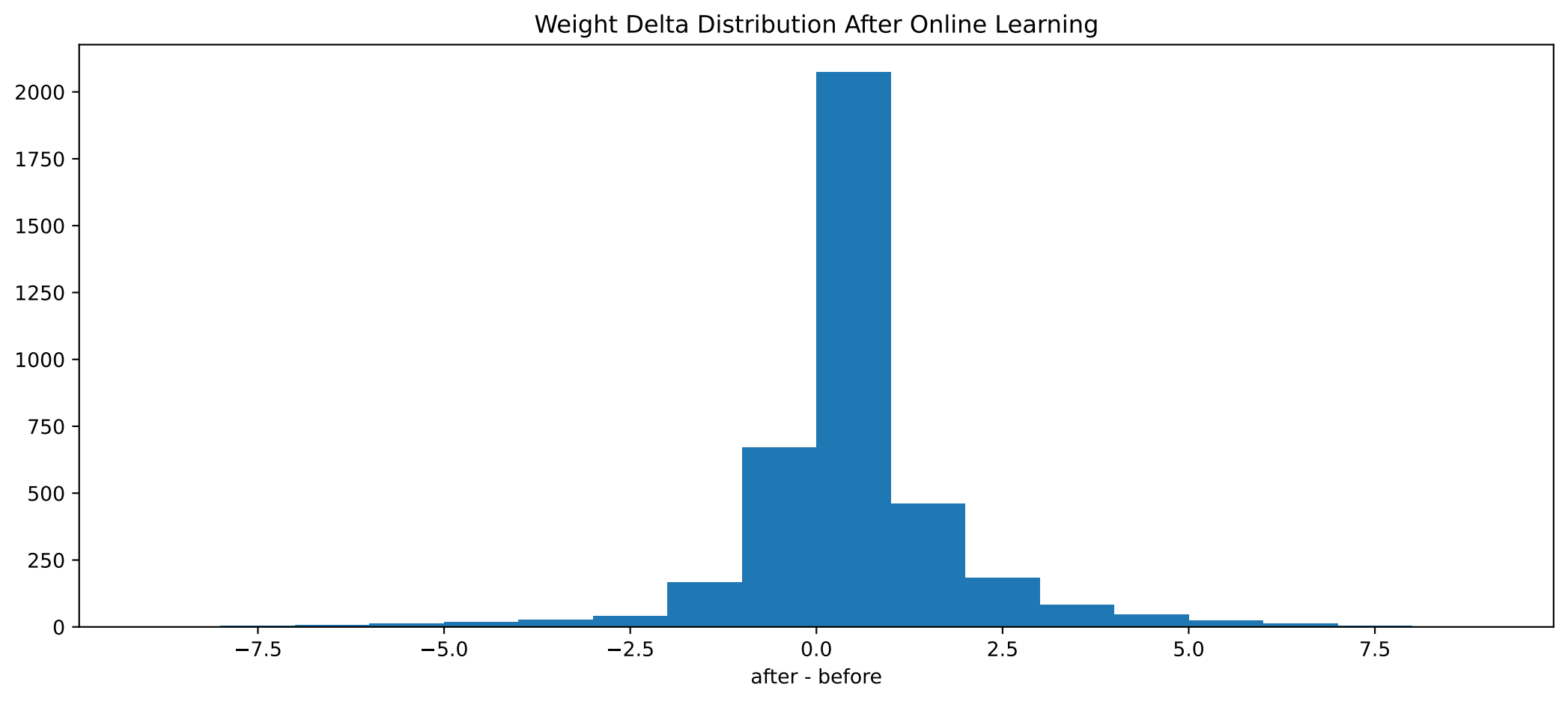

First Feature Vector Reshaped 16 x 24 For Inspection
0
2
4
6
8
10
12
14
0
5
10
15
20
5000
0
−5000
−10000

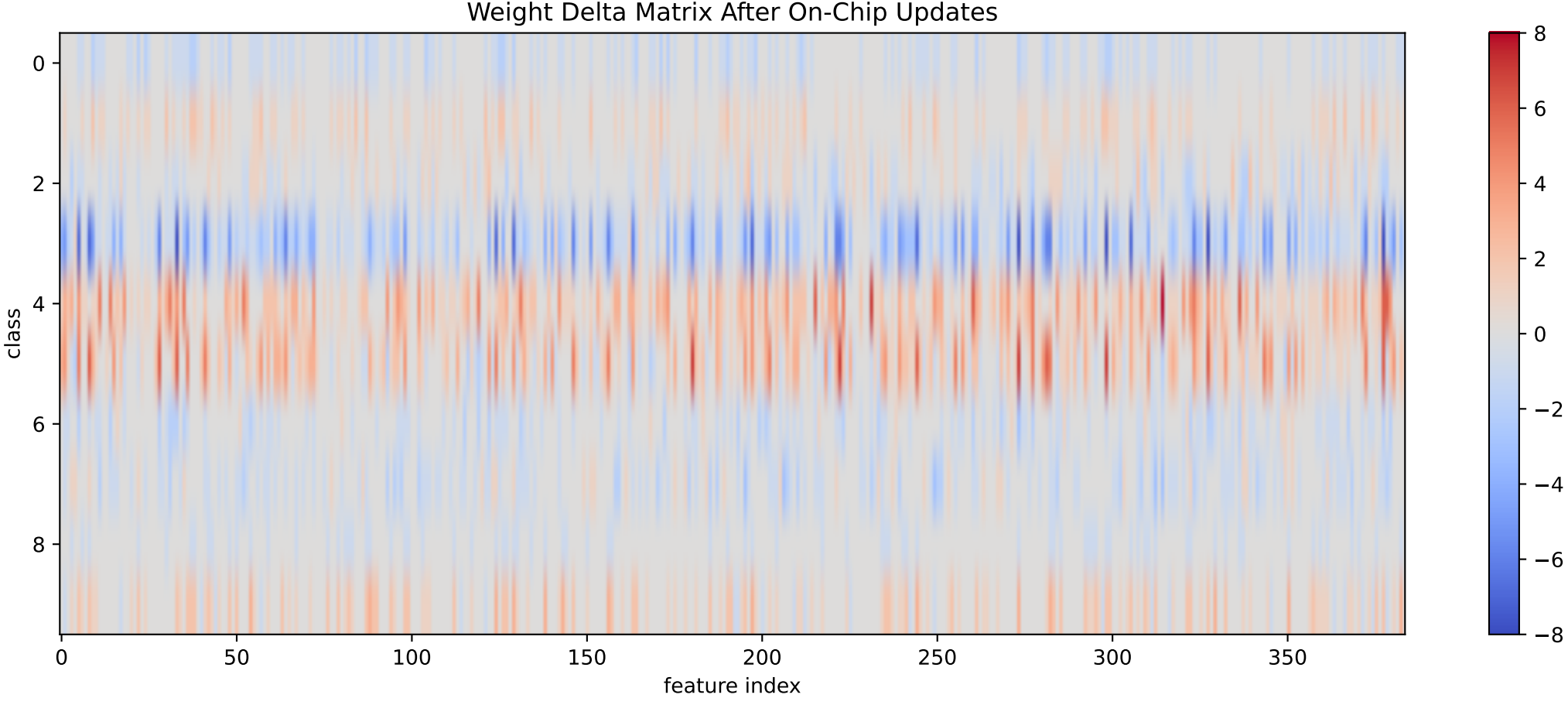

Classifier Weight Distribution Before On-Chip Updates
160
140
120
100
80
60
40
20
0
−30000
−20000
−10000
0
10000
20000
30000
int16 weight

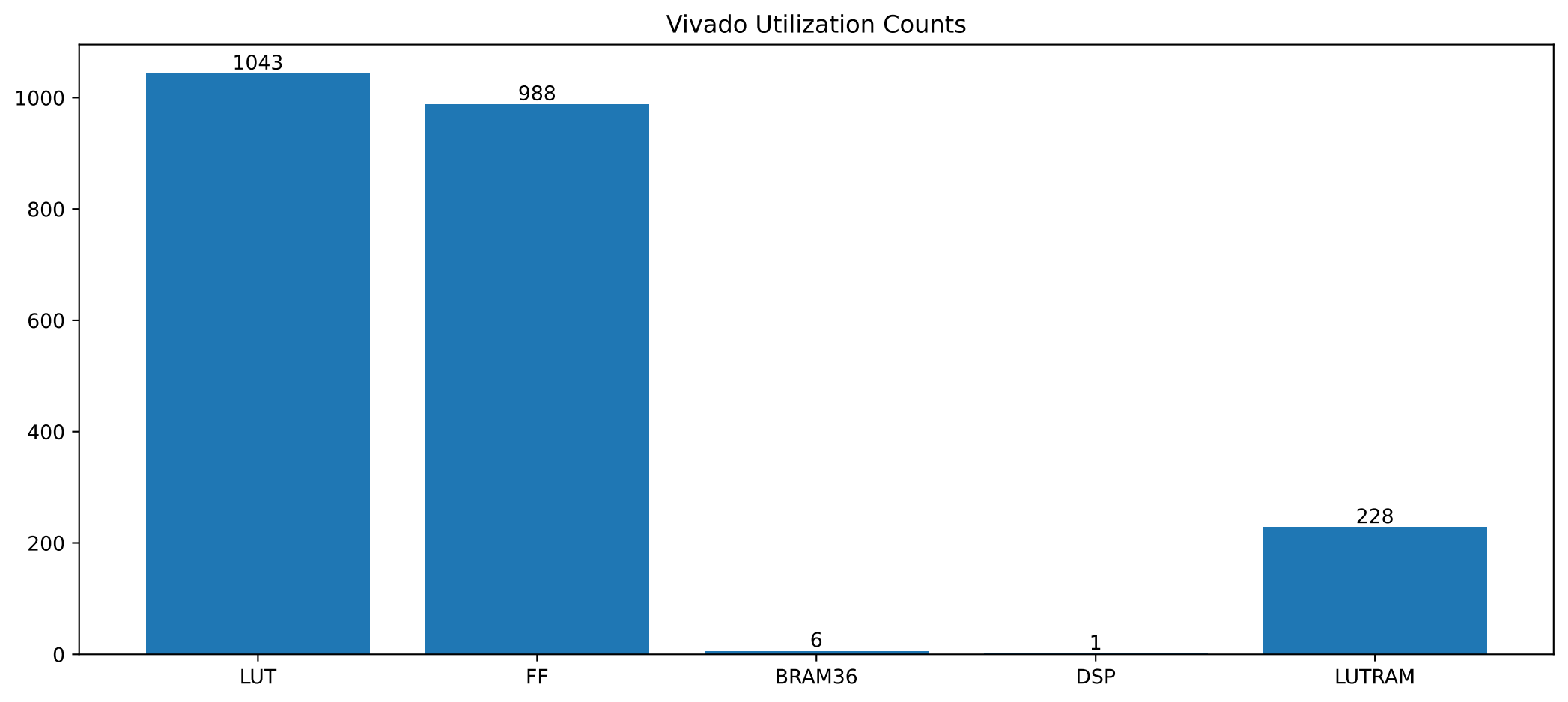

Classifier Weight Matrix: Class x Feature
class
feature index
0
2
4
6
8
0
50
100
150
200
250
300
350
30000
20000
10000
0
−10000
−20000
−30000

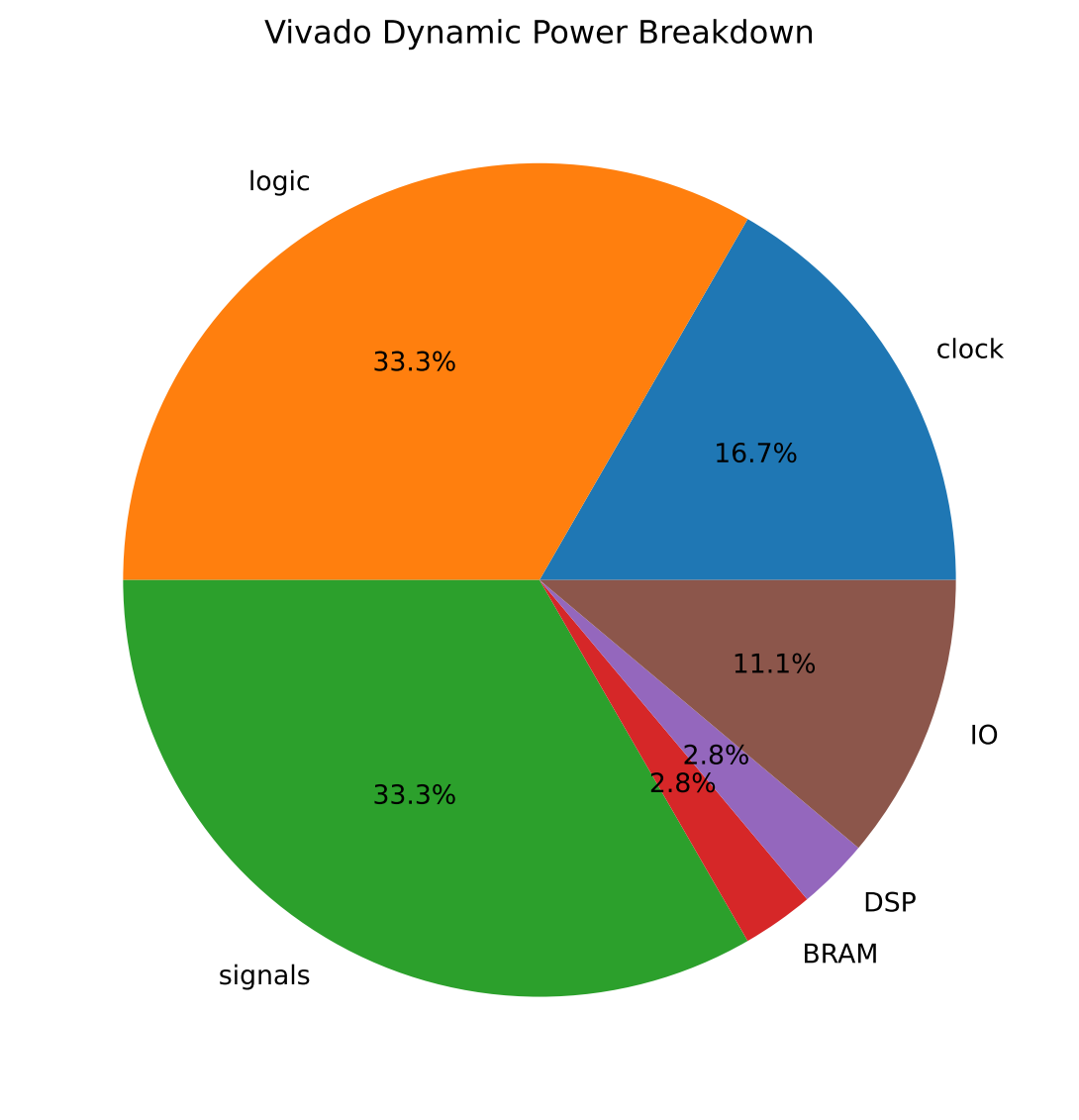

Per-Class Weight Norm
L2 norm
160000
140000
120000
100000
80000
60000
40000
20000
0
0
1
2
3
4
5
6
7
8
9

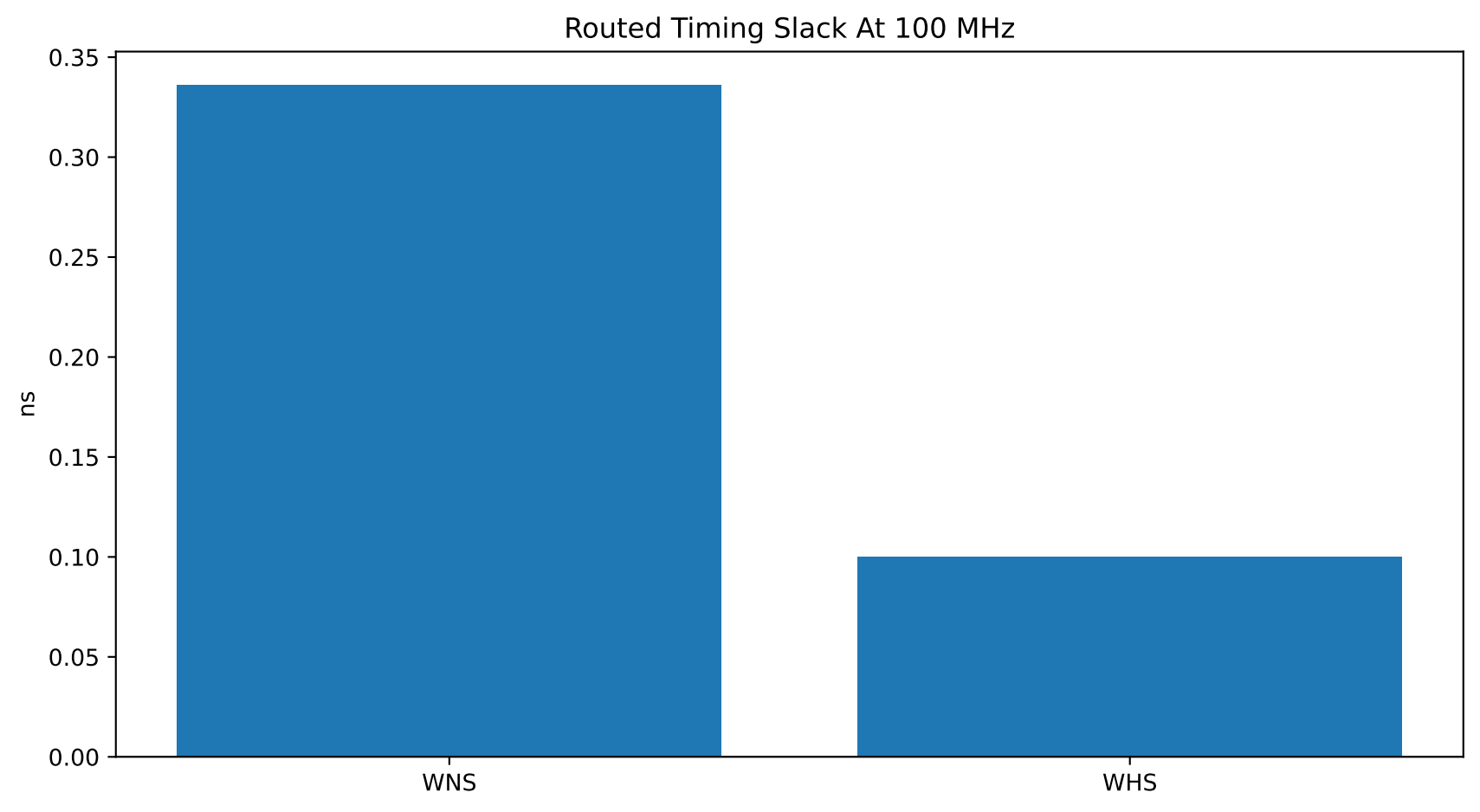

Classifier Bias Before/After On-Chip Updates
before
after
bias / 1e9
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
1
2
3
4
5
6
7
8
9


Weight Delta Distribution After Online Learning
2000
1750
1500
1250
1000
750
500
250
0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
after - before


Weight Delta Matrix After On-Chip Updates
class
feature index
0
2
4
6
8
0
50
100
150
200
250
300
350
8
6
4
2
0
−2
−4
−6
−8


Vivado Utilization Counts
1043
988
6
1
228
0
200
400
600
800
1000
LUT
FF
BRAM36
DSP
LUTRAM


Vivado Dynamic Power Breakdown
logic
33.3%
clock
16.7%
11.1%
IO
2.8%
DSP
2.8%
BRAM
33.3%
signals


Routed Timing Slack At 100 MHz
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
ns
WNS
WHS

Pipeline / Latency Accounting

Feature load: 384 accepted feature words per frame through AXI-stream wrapper.

Per class: CLASS_INIT plus one MAC_READ per feature; active features run MAC_MUL and MAC_ACCUM, while gated features jump directly to CLASS_DONE.

Activity gate: abs(feature) <= 2048 bypasses DSP multiply/accumulate for inference; measured exported-set skip fraction is 57.81%.

RTL inference skip counter 22,199,950 removes 44,399,900 MAC_MUL/MAC_ACCUM cycles over the 10k fixture.

Learning update, when applied: active features update true and predicted class rows; gated features skip the no-op read/calc/write sequence.

RTL learning skip counter 5,603 removes approximately 28,015 true/pred update cycles in the full learning run.

Learning is margin-gated, so the full 10k validation stream applied 25 updates.